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How Artificial Intelligence Could Reshape Four Essential Competitions in Future Warfare

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About This Report

How will advances in artificial intelligence (AI) shape the future of war? In this report, we offer a conceptual framework and preliminary assessment of AI's potential to disrupt how militaries fight and win wars. We use the framework to evaluate how AI could affect four “building block” competitions in military affairs. By using this building block approach, we break down warfare into component competitions revolving around (1) quantity versus quality, (2) hiding versus finding, (3) centralized versus decentralized command and control, and (4) cyber offense versus cyber defense. Our findings suggest that the U.S. military might need to change important aspects of how it traditionally operates in order to exploit AI's potential.

This report is intended for policymakers, defense analysts, and technologists considering the potential impact of AI on future military conflicts. A previous version of this report appeared as a working paper on rand.org in July 2025, titled “An AI Revolution in Military Affairs? How Artificial Intelligence Could Reshape Future Warfare.”

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Summary

Issue

How will advances in artificial intelligence (AI) shape the future of war? Despite significant public interest in AI's potential to revolutionize warfare, researchers are still in the early stages of exploring how AI will change warfighting. In this report, we offer a conceptual framework and preliminary assessment to help set the terms for a more systematic debate about AI's potential to change how militaries will fight and win future wars.

Approach

In our analysis, we assume that AI delivers on its goal of removing the limits of human intelligence as a constraint on military operations, and we examine the consequences of this assumption. We use a “building block” conceptual framework to identify essential component competitions in warfare and to evaluate how AI could affect them. We assess four competitions revolving around (1) quantity versus quality, (2) hiding versus finding, (3) centralized versus decentralized command and control (C2), and (4) cyber offense versus cyber defense. We identify ways in which AI could affect key drivers and trade-offs that will shape each building block.

Key Findings

Our analysis of the four building block competitions suggests that the U.S. military might need to change important aspects of how it traditionally operates to exploit AI's potential. Because AI technology is still maturing, our hypotheses are necessarily tentative.

- **Quantity could gain a significant edge over quality.** The relative cost-effectiveness of quantity could improve as uncrewed systems become cheaper and more capable. Advances in autonomy and robotics might make it feasible for militaries to field platforms in numbers that were previously financially impractical without dramatically sacrificing performance. The combination of more precise robotic mass and affordable robotic mass could create a cost advantage over exquisite platforms and weapons (i.e., very capable but very expensive systems), even though the latter will still have a role in future fights.
- **More-sophisticated hiding could help offset advances in finding, but this will require new approaches and investments in deception.** AI will improve finding by quickly fusing and analyzing intelligence from proliferated sensors, but analysts sometimes overstate the extent of these advantages while overlooking countermeasures that are available to the hider. Militaries can leverage AI not only for finding but also for hiding, such as by creating AI “fog of war machines” that orchestrate sophisticated deception campaigns with large numbers of autonomous decoys. Alongside other countermeasures, such as attacking the enemy's sensors, the result will be an especially contested measure-countermeasure race. The relative edge between hidiers and finders will depend on such factors as the kind of information sought, the domain and physical operating environment, and which side can better bring mass to bear in fielding sensors or decoys.
- **Mission command—a hybrid of centralized and decentralized C2 models—will remain desirable.** AI would not change the reasons mission command has advantages over more-centralized or more-decentralized C2 paradigms, which are rooted in having the right information at the right places to make certain time-sensitive decisions. Access to information, not just cognitive capacity, acts as a limiting factor.

- **Cyber defenses will benefit from AI in ways that could make battle networks more resilient against cyberattacks in the long term.** AI could help address challenges with scale, speed, and effectiveness that currently limit the cyber defense and give the cyber offense a structural edge. However, the offense will also benefit from AI, especially in the short term, and attackers will always retain some ability to penetrate networks.
- **Militaries that fail to embrace mass and deception might be at a serious disadvantage in a world of advanced AI.** Relying on small force structures of exquisite capabilities is quickly becoming a liability rather than an asset. Advances in sensing networks and long-range strike capabilities mean that militaries must place deception and counter-reconnaissance at the center of how they operate to limit their losses. Shifting to greater mass will require new approaches to U.S. defense production and sustainment.

Implications

Militaries that embrace significant changes to their operational concepts and force structures will have advantages over those that simply use AI to make their existing warfighting approaches incrementally better. Exploiting AI's military potential will be as much an organizational challenge as a technological one because it could require making disruptive changes to the ways in which the U.S. military has traditionally organized and operated.

- **The United States should begin investing now in mass and new capabilities for deception.** Opportunities to capitalize on the trends that we identify in this report are available today, even if AI is still maturing. The United States should increase its investments and experiments in key areas, such as by developing new types of attritable uncrewed systems and developing new AI tools for deception.
- **The United States needs to allocate scarce resources under the assumption that it will face sophisticated and adaptive adversaries.** There might be temptations to use AI to double down on existing approaches in the hope that the United States will have an unassailable AI first-mover advantage, allowing it to operate in ways that defy the trends we hypothesize. The United States should instead allocate scarce resources toward approaches that will provide durable long-term advantages.
- **The United States needs a plan to manage the transition to an AI-enabled force.** Important questions remain about the implementation details of leveraging AI; for example, what might a new balance between exquisite capabilities and robotic mass in the U.S. force structure look like, how can the United States field AI tools to units in a thoughtful way that balances speed and the need for trust and reliability, and how can the United States train both human personnel and AI tools in symbiotic ways that reinforce strengths and offset weaknesses?

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Introduction

How will advances in artificial intelligence (AI) influence the conduct of war? There is a growing belief among policymakers and analysts that AI will transform the future of war and could even reshape the international balance of power.¹ Despite significant public interest in AI's potential to revolutionize warfare, researchers are still in the early stages of understanding how AI will actually change warfighting.² Commentators sometimes posit that AI will abruptly lead to superintelligence and superweapons without fully considering the broader military context in which these new technologies would operate. In this report, we propose a conceptual framework to help set the terms for a more systematic debate about AI's implications for the future of war by breaking down the broad question of how AI will shape warfare into discrete "building blocks" that are foundational to modern military operations. We then provide a preliminary assessment of how AI could change the dynamics in four critical building block competitions. We predict that the characteristics of AI-enabled warfare will reward militaries that invest in capabilities and operational concepts that leverage mass, deception, mission command, and cybersecurity.

Assumptions About AI

Because AI is still an emerging technology, our assessment about how AI could shape military operations is necessarily tentative and rests on assumptions about the quality and breadth of its future performance. Militaries already apply artificial narrow intelligence (ANI) to discrete tasks under specific conditions, and ANI sometimes exceeds human performance. In the future, artificial general intelligence (AGI) might consistently match human performance across a wide variety of cognitive tasks and do so reliably under diverse

¹ See, for example, John A. Tirpak, "Kendall: In US-China 'Race for Technological Superiority,' AI May Be the Key," *Air & Space Forces*, October 29, 2024; Michèle Flournoy, "AI Is Already at War: How Artificial Intelligence Will Transform the Military," *Foreign Affairs*, Vol. 102, No. 6, November/December 2023; and "AI Will Transform the Character of Warfare," *The Economist*, June 20, 2024. On the potential implications for the balance of power, see Michael C. Horowitz, "Artificial Intelligence, International Competition, and the Balance of Power," *Texas National Security Review*, Vol. 1, No. 3, May 2018; and Elliot Ackerman and James Stavridis, "Drone Swarms Are About to Change the Balance of Military Power," *Wall Street Journal*, March 14, 2024.

² For examples of earlier work on this topic, see Paul Scharre, *Army of None: Autonomous Weapons and the Future of War*, W. W. Norton & Company, 2018; Paul Scharre, *Four Battlegrounds: Power in the Age of Artificial Intelligence*, W. W. Norton & Company, 2023a; Avi Goldfarb and Jon R. Lindsay, "Prediction and Judgment: Why Artificial Intelligence Increases the Importance of Humans in War," *International Security*, Vol. 46, No. 3, Winter 2021/22; Benjamin Angel Chang, *Artificial Intelligence and the US-China Balance of Power*, dissertation, Massachusetts Institute of Technology, 2021; Lance Menthe, Li Ang Zhang, Edward Geist, Joshua Steier, Aaron B. Frank, Erik Van Hegewald, Gary J. Briggs, Keller Scholl, Yusuf Ashpari, and Anthony Jacques, *Understanding the Limits of Artificial Intelligence for Warfighters: Vol. 1, Summary*, RAND Corporation, RR-A1722-1, 2024; Benjamin Jensen, Christopher Whyte, and Scott Cuomo, "Algorithms at War: The Promise, Peril, and Limits of Artificial Intelligence," *International Studies Review*, Vol. 22, No. 3, September 2020; Kenneth Payne, *I, Warbot: The Dawn of Artificially Intelligent Conflict*, Oxford University Press, 2021; and Anthony King, *AI, Automation, and War: The Rise of a Military-Tech Complex*, Princeton University Press, 2025.

conditions. This report focuses on where AI might be headed rather than where the technology is today. Our analysis explores the implications for warfare if AI can offer human-level performance for any cognitive task that is relevant to military operations, either because of breakthroughs in AGI or because militaries develop sophisticated ANI tools for many different applications. To guide our analysis, we used the heuristic that AI might *remove the limits of human cognitive capacity as a constraint* on military operations. For example, we assume that AI could start filling roles that only humans were once capable of, such as flying aircraft or writing computer code, and we assume that AI could do so at comparable levels of performance to humans. We do *not* assume that AI becomes superintelligent in the sense that it always outperforms the best humans in every role.

We assume that, as with all technologies, even AGI would face limitations in what it could do. To avoid treating AI as a magic wand able to solve all military problems, we assume that the laws of physics and information theory (as they are currently understood) will continue to hold.³ This assumption means, for example, that AI will not lead to the development of aircraft with limitless energy that do not contend with such mundane factors as fuel or range constraints. Other factors—such as size, weight, and power constraints—will still limit military operations even if human cognitive capacity does not. The assumption also means that AI cannot make decisions using information it does not have. AI or AI-enhanced reasoning capabilities run into limits if the information necessary to make a better decision is simply not available.

How AI will affect military operations depends on its integration with other technologies. AI is essentially software that can substitute for cognitive capacity, but substituting for human physical capacity and interacting with the physical world requires combining AI with hardware—especially robotic systems. Militaries are already integrating AI into some robotic systems, such as uncrewed aerial vehicles. Research to improve robotic systems is still ongoing, and humanoid robots still lag far behind progress in uncrewed vehicles.⁴ We do not make strong assumptions about what the state of the art looks like for these supporting technologies, whether for robotics or for other areas, such as directed energy and quantum sensing.⁵ When relevant, we discuss how different assumptions about future developments in other technologies might influence our conclusions.

Mechanisms Through Which AI Could Change Warfare

Understanding AI's potential to reshape warfare requires identifying specific mechanisms through which the technology could change how militaries operate. One of AI's defining features as a class of technology is that it is a "general-purpose technology" rather than one discrete capability. AI has many different potential applications, and most of its effects occur through its interactions with a wide variety of other established

³ It is theoretically possible that AI might accelerate research that fundamentally changes our understanding of physics, but such scenarios are beyond the scope of this report; see Edward Geist and Alvin Moon, *What Even Superintelligent Computers Can't Do: A Preliminary Framework for Identifying Fundamental Limits Constraining Artificial General Intelligence*, RAND Corporation, WR-A3990-1, 2025.

⁴ Hannah Miao and Yoko Kubota, "Before They Can Rule the World, Chinese Robots Need to Master Basic Chores," *Wall Street Journal*, August 16, 2025.

⁵ Because our analysis explores the implications of advanced AI in a world in which other technologies continue to develop, an implicit assumption is that advanced AI will be available to militaries reasonably soon (as of this writing in late 2025). Although we do not assume a specific timeline, our analysis is more applicable to changes over the next ten to 20 years rather than the next 50 to 100 years because, over the course of the latter time frame, there might be many other major technological and political changes that could make this report's analysis obsolete.

and emerging technologies.⁶ This makes AI more like electricity or the internal combustion engine than a specific type of platform or weapon, such as tanks, aircraft, or nuclear weapons.⁷ Because AI is an enabling technology with many applications, and because different militaries will adopt AI for a wide variety of purposes to meet their particular needs, there is no *single* way that the technology will change warfare.⁸ AI's effects are also likely to vary across actors and in different settings. This section therefore outlines in broad terms a variety of ways that militaries might leverage AI.

In describing AI's applications, we distinguish between warfighting and preparing for war.⁹ *Warfighting* is the realm of active combat operations, characterized by war's dynamic and unpredictable nature, in which the fog of war, friction, and complex interactions with the enemy require militaries to be flexible and adaptable. AI could change warfighting through mechanisms that include the following:¹⁰

- **Insight** (analyzing information and generating knowledge). By synthesizing and interpreting vast amounts of data, AI can improve operators' understanding of the battlespace. Many of AI's potential cognitive benefits do not depend on AI surpassing human levels of intelligence as much as the ability to reason about vastly more data on a much faster timescale. AI can provide insight on a wide variety of issues, from intelligence analysis that helps close kill chains to analysis of logistical requirements. As a tactical example, the U.S. military deployed an AI-enabled image classifier as early as 2017 to sort through huge volumes of intelligence and detect specific objects, such as vehicles, in support of operations against the Islamic State.¹¹ As a logistical example, testing AI for predictive maintenance increased mission-capable rates for U.S. aircraft.¹²
- **Autonomy** (replacing human labor). Just as AI might replace the cognitive labor of workers in the civilian economy, it might automate cognitive work in the military. When paired with robotic systems, AI can also automate physical work. Labor automation offers the potential for better performance, reduced casualties, and reduced personnel costs. As an example, in 2024, the U.S. Air Force (USAF) announced that an AI-controlled F-16 fighter put up a "roughly even fight" in an exercise against a crewed F-16 with an expert human pilot.¹³ Replacing human labor could be particularly useful for tasks that are constrained by human physical limitations, including piloting aircraft on missions requiring very long endurance or involving high g-force loads. Replacing onboard pilots with AI could also enable air-

⁶ This is true for AI as a class of technology, regardless of whether any individual AI tool's breadth is narrow or general.

⁷ Horowitz, 2018, p. 41.

⁸ Michael C. Horowitz, *The Diffusion of Military Power: Causes and Consequences for International Politics*, Princeton University Press, 2010.

⁹ Similarly, Colin Gray draws a distinction between *war proper*—i.e., warfighting—and *preparation for war* (Colin S. Gray, *Modern Strategy*, Oxford University Press, 1999). Michael Howard also distinguishes between the operational (war proper) and logistical (preparation for war) dimensions of strategy (Michael Howard, "The Forgotten Dimensions of Strategy," *Foreign Affairs*, Vol. 57, No. 5, Summer 1979).

¹⁰ Karl Mueller originally developed this framework inductively by surveying potential applications of AI. The list is not exhaustive, but it does include key applications. We later applied the terminology of *insight*, *autonomy*, and *decision support* from Shashank Joshi's work (Shashank Joshi, "AI on the Battlefield: A Fait Accompli?" *Survival*, Vol. 67, No. 2, 2025). See also Forrest E. Morgan, Benjamin Boudreaux, Andrew J. Lohn, Mark Ashby, Christian Curriden, Kelly Klima, and Derek Grossman, *Military Applications of Artificial Intelligence: Ethical Concerns in an Uncertain World*, RAND Corporation, RR-3139-1-AF, 2020.

¹¹ Scharre, 2023a, pp. 56–58.

¹² Scharre, 2023a, pp. 196–197.

¹³ Greg Hadley, "In F-16 Dogfight, AI and Human Pilots Are 'Roughly an Even Fight,' Says Kendall," *Air & Space Forces*, May 8, 2024.

frames to be smaller and to be produced in larger quantities without the need to train a correspondingly greater number of crews.

- **Management** (coordinating complicated actions and systems). AI could manage complicated tasks that involve too many entities or components for humans to control at sufficient speed and scale to be effective.¹⁴ Whereas the autonomy mechanism described above involves AI performing tasks that humans could do, management is about AI performing tasks that exceed human capacity. For example, researchers are developing AI tools to manage large drone swarms that would be impractical for humans to pilot on a one-to-one basis.¹⁵ AI might make the challenges of deconflicting crowded airspace or adaptively synchronizing the fires of hundreds of kinetic and nonkinetic weapons manageable for machines in circumstances that would be too complex for humans.
- **Decision support** (improving decisionmaking). AI could provide decision support tools to help military and civilian decisionmakers better assess their strategic options, evaluate different courses of action (COAs), and translate decisions into orders and coordinating guidance.¹⁶ AI is already better than humans at playing well-structured strategy games, such as chess and Go, and human players can reach peak performance when working in concert with AI tools.¹⁷ Future advances in AI could extend the decisionmaking benefits of human-machine teaming to more-ambiguous and more-dynamic contexts, such as war. Advances in AI also have the potential to enable humans to make faster decisions, potentially providing an advantage over opponents when rapid decisionmaking and quick responses to an adversary's actions are important.¹⁸

Beyond combat operations, AI could change how countries prepare for war and sustain their operations. Preparation for war encompasses such activities as training and force development, technological innovation, materiel production, and rear-area logistics. These support functions are relevant for both building military capabilities during peacetime and sustaining forces during wartime. The same mechanisms described above could affect preparation for war:

- **Insight.** AI could help design new military capabilities by spurring technological discoveries and advancing military research and development efforts.¹⁹ As part of an emerging “AI for Science” paradigm, researchers in many fields are integrating AI into their work processes to speed up the pace of discovery.²⁰ Scientists have used AI to achieve Nobel Prize-winning breakthroughs in chemistry, and

¹⁴ Matthew Walsh, Lance Menhe, Edward Geist, Eric Hastings, Joshua Kerrigan, Jasmin Léveillé, Joshua Margolis, Nicholas Martin, and Brian P. Donnelly, *Exploring the Feasibility and Utility of Machine Learning-Assisted Command and Control: Vol. 1, Findings and Recommendations*, RAND Corporation, RR-A263-1, 2021.

¹⁵ Scharre, 2018, p. 11.

¹⁶ Eric Robinson, Daniel Egel, and George Bailey, *Machine Learning for Operational Decisionmaking in Competition and Conflict: A Demonstration Using the Conflict in Eastern Ukraine*, RAND Corporation, RR-A815-1, 2023; David Stebbins, Richard S. Girven, Timothy Parker, Thomas Deen, Brandon de Bruhl, James Ryseff, Jessica Welburn Paige, Annie Yu Kleiman, Sunny D. Bhatt, Éder M. Sousa, Marta Kepe, and Matthew Fay, *Exploring Artificial Intelligence Use to Mitigate Potential Human Bias Within U.S. Army Intelligence Preparation of the Battlefield Process*, RAND Corporation, RR-A2763-1, 2024.

¹⁷ Scharre, 2018, p. 321.

¹⁸ Bryan Clark, Dan Patt, and Harrison Schramm, *Mosaic Warfare: Exploiting Artificial Intelligence and Autonomous Systems to Implement Decision-Centric Operations*, Center for Strategic and Budgetary Assessments, 2020.

¹⁹ Gaurav Sett, “How AI Can Automate AI Research and Development,” commentary, RAND Corporation, October 24, 2024.

²⁰ Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, et al., “Scientific Discovery in the Age of Artificial Intelligence,” *Nature*, Vol. 620, August 30, 2023.

AI could assist with a wide variety of applied science and engineering problems in the future, including the design of software, platforms, weapons, and structural materials.²¹

- **Autonomy.** Militaries are already using AI to enhance human productivity in a variety of military administrative functions.²² When integrated with robots, AI could directly replace human labor in building and maintaining military platforms and weapons. Researchers are exploring ways to use AI for end-to-end robotic manufacturing, including making robotic manufacturing tools more adaptive so that they can work with small electronics, such as those used in drones and munitions.²³ Skilled human labor is often a bottleneck in scaling up production and dealing with maintenance backlogs, and AI could replicate the knowledge of highly skilled workers at a much lower marginal cost than that of training and retaining human personnel.
- **Management.** Building and maintaining a force structure with many more platforms than exist today will require managing complex supply chains, logistics, and maintenance at scale, particularly if current trends toward more-dispersed operations continue.²⁴ AI could help coordinate the backend operations required to field and maintain this force structure. AI could affect multiple elements of the production chain, including supply chain management, asset management, quality control, and automated production.²⁵
- **Decision support.** AI could also help optimize decisions about how to design, develop, acquire, and maintain new military capabilities. AI could aid human decisionmakers in assessing trade-offs and opportunities for how to implement targeted reforms that improve performance and reduce costs.²⁶

Table 1.1 presents a summary, using the framework presented above, of the ways in which AI could affect warfighting and preparing for war.

A Framework for Analysis: Building Block Competitions

Our goal in this report is to help U.S. defense analysts and national security practitioners better understand how AI could influence future military operations.²⁷ Such understanding can help focus attention and investment on what kinds of new capabilities and operational concepts to consider. A comprehensive analysis of AI's impact on the future of warfare in a single report is impossible because AI is a general-purpose technology that will influence how militaries operate in countless ways. However, we aim to provide greater focus to debates in select areas that deserve significant attention. Because wars are relative contests, knowing that AI could uplift almost all capabilities in absolute terms does not provide guidance about what militaries should prioritize.

²¹ Steve Lohr, "The Quest for A.I. 'Scientific Superintelligence,'" *New York Times*, March 10, 2025.

²² George I. Seffers, "U.S. Navy Going Big on AI for an Array of Tasks," *Signal*, January 30, 2025.

²³ Patrick Tucker, "Could This Breakthrough in Robot Manufacturing Reshape Warfare?" *Defense One*, April 14, 2025.

²⁴ Miranda Priebe, Alan J. Vick, Jacob L. Heim, and Meagan L. Smith, *Distributed Operations in a Contested Environment: Implications for USAF Force Presentation*, RAND Corporation, RR-2959-AF, 2019.

²⁵ Henri Bristol, Enno de Boer, Dinu de Kroon, Forest Hou, Rahul Shahani, and Federico Torti, "How Manufacturing's Lighthouses Are Capturing the Full Value of AI," McKinsey, April 9, 2024.

²⁶ Li Ang Zhang, Yusuf Ashpari, and Anthony Jacques, *Understanding the Limits of Artificial Intelligence for Warfighters: Vol. 3, Predictive Maintenance*, RAND Corporation, RR-A1722-3, 2024.

²⁷ Our analysis is intended to inform decisions for the U.S. military. Other militaries might have different political, organizational, and operational backgrounds that change the approaches that make sense for their specific contexts.

TABLE 1.1
Ways That AI Could Affect Warfare

Mechanism	Logic	Example Use Cases
Warfighting		
Insight (Analyzing information and generating knowledge)	AI makes it possible to analyze much larger volumes of data and generate information at much faster speeds than are practical for humans.	• Multisource sensor fusion • Targeting and surveillance • Cyber operations
Autonomy (Replacing human labor)	AI can make autonomous systems more capable of replacing human labor for cognitive tasks and, when combined with robotics, for physical tasks.	• Piloting aircraft and other vehicles • Supply convoys • Maintenance operations
Management (Managing complicated actions and systems)	AI can coordinate and synchronize activities that are too complex for humans to manage at tactically or operationally relevant timescales.	• Managing intelligent robotic swarms • Coordinating airspace for thousands of platforms • Synchronizing kinetic and nonkinetic fires at scale
Decision support (Improving decisionmaking)	AI can help humans make faster and potentially better decisions than humans could alone because of AI's pattern recognition capabilities and less-constrained thinking.	• Generating and evaluating COAs • Translating decisions into more-actionable and more-specific guidance
Preparing for war		
Insight	AI can help humans design new military capabilities by accelerating research and development.	• Improved research in capabilities, such as directed-energy weapons, and advanced structural material performance (e.g., engines, airframes, armor)
Autonomy	Combined with investments in robotics, AI can help replace skilled human labor in building and maintaining platforms and weapons.	• Automated production lines • Robotic maintenance depots
Management	AI can help humans manage complex supply chain, logistics, and maintenance operations.	• Managing more-dynamic logistical operations to sustain a dispersed force structure with potentially thousands of uncrewed platforms operating across multiple domains
Decision support	AI can help humans optimize production, logistics, and force design.	• AI-assisted expertise in optimization to balance such considerations as performance, costs, and speed

SOURCE: Features the authors' analysis and insights from Joshi, 2025.

NOTE: These findings reflect assumptions about AI's potential capabilities using first principles rather than research findings based on technical analysis or modeling.

One framework for guiding choices about priorities is to look at building block competitions in military affairs. These building blocks are fundamental military-technological competitions that are critical to how militaries fight and win wars.²⁸ If AI is going to cause a broader revolution in warfare, it will rest on the microfoundations of how AI changes specific ways that militaries operate. Building blocks provide a framework for analyzing what changes in relative advantages might look like.

²⁸ For another example of the building block approach, see Andrew F. Krepinevich, *The Origins of Victory: How Disruptive Military Innovation Determines the Fates of Great Powers*, Yale University Press, 2023.

Building block competitions highlight choices and trade-offs. Militaries must prioritize certain ways of operating over others.²⁹ We focus on four building block competitions in this report. First, we look at quantity versus quality—what is the relative advantage of having more forces or better forces? Second, hiding versus finding—what is the relative advantage between scouting and reconnaissance and deception and counter-reconnaissance? Third, centralized versus decentralized command and control (C2)—what is the relative advantage between consolidating decisionmaking at higher echelons and delegating it to forces closer to the tactical edge? Fourth, cyber offense versus cyber defense—what is the relative advantage between attacking computer networks and defending them?

We present these building block competitions in binary terms to identify clear trends in the direction of how AI might shift areas of relative advantage toward one end of the spectrum or the other in each competition. This does not mean that the U.S. military faces all-or-nothing choices in these competitions. For example, regarding the quantity-versus-quality building block competition, AI and robotics are already combining in ways that reinforce the advantages of lower-cost uncrewed systems relative to exquisite platforms, but this does not mean that exquisite systems have no place in future battlefields or that the U.S. military should only invest in cheap drones. The core argument is that the cost-effectiveness of quantity versus that of quality is changing, and the U.S. military should reconsider the extent of its emphasis on exquisite capabilities and how to leverage robotic mass. Although the U.S. military cannot focus on only one end of the spectrum while ignoring the other, the U.S. military needs to set priorities and make bets on how AI will empower certain approaches over others because resources are finite. Investing a marginal dollar toward a larger U.S. force structure would compete with investing in qualitatively better capabilities. Given the magnitude of the military challenge that China poses, the United States can no longer pursue overmatch across the board; instead, it must establish priorities while guarding against spreading its resources too thin by hedging on everything.³⁰

We chose these building blocks because we believe that they are central to modern military operations and that they will remain central in the future as technology evolves. Modern *battle networks* integrate (1) sensors that observe the battlespace and find targets (e.g., reconnaissance drones or imaging satellites), (2) shooters that generate kinetic and nonkinetic effects (e.g., cruise missiles or cyberattacks), (3) communications links that enable critical information to flow through the network, and (4) C2 elements to make sound and timely decisions using that information.³¹ Even in a world with advanced AI, the U.S. military will still grapple with familiar questions, such as how to scout for enemy forces while protecting its own forces from enemy sensors and how to protect the computer networks that provide the electronic backbone of its operations. Some questions that appear new, such as how to integrate a new generation of drones into its force structure and how to leverage AI as a decisionmaking tool, still evoke classic debates in defense planning, such as how to strike the balance between quantity versus quality and how to manage the trade-offs of different ways of exercising C2 over fielded forces.

²⁹ The competition aspect of the building blocks refers to the choice of certain approaches over others, such as prioritizing quantity over quality. It is not referring to a competition with an adversary. Both the United States and China, for example, could choose to prioritize quantity over quality at the same time.

³⁰ Eric Heginbotham, Michael Nixon, Forrest E. Morgan, Jacob L. Heim, Jeff Hagen, Sheng Tao Li, Jeffrey Engstrom, Martin C. Libicki, Paul DeLuca, David A. Shlapak, David R. Frelinger, Burgess Laird, Kyle Brady, and Lyle J. Morris, *The U.S.-China Military Scorecard: Forces, Geography, and the Evolving Balance of Power, 1996–2017*, RAND Corporation, RR-392-AF, 2015; U.S. Department of Defense, *Military and Security Developments Involving the People's Republic of China*, 2024.

³¹ Todd Harrison, *Battle Networks and the Future Force, Part 1: A Framework for Debate*, Center for Strategic and International Studies, August 2021; John Stillion and Bryan Clark, *What It Takes to Win: Succeeding in 21st Century Battle Network Competitions*, Center for Strategic and Budgetary Assessments, 2015.

We offer predictions for how AI will shape relative advantages in the four building block competitions. Table 1.2 summarizes these predictions; we also refer to these predictions as *hypotheses* throughout the report. Because AI is an enabling technology that militaries will deploy in applied settings with specific operational requirements and in interaction with other technologies, it is implausible to make these predictions based solely on the nature of the technology itself. We therefore consider how our guiding heuristic—removing human cognitive capacity as a constraint on military operations—would influence the key drivers and trade-offs that already shape each building block competition. We explain the logic of the hypotheses in detail, as well as their scope conditions and the conditions under which they might be wrong, in the individual chapters that follow. Because AI is an emerging technology and we are projecting years into the future, these are necessarily tentative predictions.

An important scope condition is that these predictions are about trends in long-term competition between peers, in which both sides are technologically sophisticated and adaptive. We assume that neither side enjoys an overwhelming advantage in resources or vastly superior technology (especially a monopoly on advanced AI) and that both sides will seek to develop countermeasures against their rivals’ strategies, capabilities, and concepts of operations.³² Militaries must therefore make bets on strategies, concepts, and capabilities knowing that a competitor might employ AI-enabled countermeasures. Militaries might still operate in ways that defy long-term trends, but they will ultimately pay a price for choosing to do so. This suggests that the United States should embrace “no regrets” adaptations rather than investing in approaches that make sense only during a transition in which the United States might have a short-lived first-mover advantage in advanced AI. Assessing the consequences of a first-mover advantage in detail and the conditions under which it would still be prudent to seize short-term opportunities even if they are fleeting is beyond this report’s scope.

TABLE 1.2
Predictions About the Long-Term Military Implications of Advanced AI

Building Block Competition	Hypothesized Advantage	Logic
Quality versus quantity	Quantity	AI will enable militaries to field large numbers of “good-enough” autonomous uncrewed systems that can offer a cost advantage over relying on small numbers of expensive, exquisite systems.
Hider versus finder	Contested	AI will enable more-effective surveillance through improved sensor fusion and fielding autonomous sensing systems. But AI can also enable better deception to offset the finder’s gains. This requires investing in such countermeasures as “fog of war machines,” which use AI to orchestrate robotic decoys and deceptive tactics that increase the finder’s uncertainty. Whether the finder or the hider will have an edge will depend heavily on the context, including the type of information sought, the physical operating environment, and whether the finder or the hider has advantages in mass.
Centralized versus decentralized C2	Mission command	AI does not change the reasons that mission command has advantages over more-centralized or more-decentralized C2 paradigms, which are rooted in having access to the right information at the right times and places rather than in cognitive capacity alone. AI will improve C2 within the mission command paradigm rather than replace it.
Cyber offense versus cyber defense	Cyber defense	AI will facilitate stronger cyber defenses in the long term by addressing some of the limitations of speed, scale, and effectiveness that currently hold the defender back and give the attacker a relative edge.

³² This follows Andrew Krepinevich’s framing of a “mature precision-strike regime” but in the context of AI (Andrew F. Krepinevich, *Maritime Competition in a Mature Precision-Strike Regime*, Center for Strategic and Budgetary Assessments, April 13, 2015). See also Stephen Biddle and Ivan Oelrich, “Future Warfare in the Western Pacific: Chinese Antiaccess/Area Denial, U.S. AirSea Battle, and Command of the Commons in East Asia,” *International Security*, Vol. 41, No. 1, Summer 2016.

These building block competitions are not the only dimensions in which AI will influence military operations, and future research should extend this scope to include others. Potential candidates include competing approaches to logistics or trends in air and missile threats versus defenses against them. Future research should also unpack the implementation details of the trends we hypothesize. For example, if quantity (mass) has a growing advantage over quality, what might specific U.S. force structure options look like that strike a new balance between exquisite and attritable capabilities? If the cost-effectiveness of cyber defense is improving, what does that mean for the portfolio of investments the U.S. military should make to support its needs for both offensive and defensive cyber operations? We view the four building block competitions in this report as part of an exploratory analysis that can help set the terms for a more systematic debate about AI's consequences for the U.S. military and the future of war—not the final word in that debate.

The remainder of this report proceeds as follows. In the next four chapters, we explore the four operational building block competitions. In the conclusion, we revisit the implications for the U.S. military and discuss remaining uncertainties and directions for future research.

Quality Versus Quantity

How will advances in AI influence the relative advantage between quantity and quality in military operations? A recurring theme in military history is that new technologies have sometimes allowed one side to offset a numerical disadvantage by creating a qualitative edge, such as U.S. breakthroughs in stealth and precision strike networks during the “Second Offset” in the latter half of the Cold War.¹ The U.S. military’s force planning in recent decades has relied heavily on building this type of qualitative edge by fielding smaller numbers of increasingly capable exquisite platforms. The extent of this emphasis on quality has raised periodic concerns that the U.S. military might struggle against an adversary, such as China, that prioritizes mass in its own force structure.²

We argue that AI could produce greater relative advantages for quantity over quality. This stands in contrast to many technological innovations in the past, which tended to benefit quality. There are two key elements to this prediction. The first is that advances in AI and robotics are improving the performance of cheaper uncrewed systems. Robotic mass can offer good-enough performance to meet some mission requirements without needing to match exquisite performance. In the context of precision strike, Michael Horowitz calls this trend “precise mass.” Cheaper one-way attack drones now offer a level of precision previously reserved for exquisite missiles and strike aircraft.³ This trend toward lower-cost but good-enough robotic systems will apply to other applications as well.

The second element of our argument is that robotic mass will increasingly offer a cost advantage over exquisite systems, reflecting the trend toward “affordable mass.”⁴ In the past, pursuing quantitative superiority might have been desirable in theory but cost-ineffective in practice because it required making large sacrifices in qualitative performance to afford a much bigger force structure, potentially making the total cost required to achieve the same effect higher than for using exquisite systems. In the future, the combination of precise mass and affordable mass through using AI-enabled robotic systems might change the underlying cost-effectiveness considerations of the quantity-quality competition.

Because AI will enable advances in both quantity and quality, our key analytical challenge was developing a framework to adjudicate which side of the ledger will benefit more. The goal was to evaluate the claim that AI could give quantity an edge over quality in a principled way that produced clear predictions, identified

¹ Krepinevich, 2023.

² See, for example, Mackenzie Eaglen, “The Bias for Capability over Capacity Has Created a Brittle Force,” *War on the Rocks*, November 17, 2022. The defense reform movement of the 1970s and early 1980s criticized the U.S. Department of Defense’s proclivity for smaller numbers of complex and expensive platforms over larger quantities of less complex and less expensive platforms. For a summary, see John T. Correll, “The Reformers,” *Air Force Magazine*, February 2008. For a foundational source that raised the public profile of these arguments, see James Fallows, *National Defense*, Random House, 1981.

³ Michael C. Horowitz, “Battles of Precise Mass: Technology Is Remaking War—and America Must Adapt,” *Foreign Affairs*, Vol. 103, No. 6, November/December 2024.

⁴ See, for example, Marcel Plichta and Ash Rossiter, “A One-Way Attack Drone Revolution? Affordable Mass Precision in Modern Conflict,” *Journal of Strategic Studies*, Vol. 47, No. 6–7, 2024.

conditions under which the hypothesis is wrong, and enabled other researchers to replicate the analysis. We used Lanchester equations as the framework because the equations provide a clearly structured and well-established method for exploring trade-offs between quantity and quality.⁵ The Lanchester equations have limitations, which we discuss below. We applied the Lanchester equations in the context of air-to-air combat. To analyze changes in cost dynamics, we also conducted a basic costing analysis that leveraged classic cost-estimating relationships that use notional aircraft attributes to estimate aircraft cost.

The analysis suggests that AI will favor quantity over quality if AI can help make robotic mass not only more capable but also cheaper. AI is unlikely to offer large-enough qualitative advantages for a small number of exquisite forces to counterbalance a mass of good-enough forces while retaining a cost advantage.

How AI Can Help Both Quantity and Quality in Air Combat

The war in Ukraine has demonstrated the potential for militaries to generate robotic mass on a scale that was previously difficult to imagine. But many of the drones deployed in Ukraine are basic systems, and they are often vulnerable to basic attacks. By some estimates, Ukraine lost as many as 10,000 drones per month in 2023 because Russian forces jammed the radio-frequency links required for Ukraine's operators to control the drones.⁶ Advances in AI, however, could make it possible for drones to operate effectively without human control. In this chapter, we shift the focus from the first-person-view drones that are common in Ukraine to more-advanced uncrewed fighter aircraft that could be important for air-to-air combat in future conflicts.

AI could enable militaries to significantly increase the number of combat aircraft in their force structures by reducing costs in several ways. Fielding AI-enabled autonomous aircraft could enable aircraft to be smaller, lighter, and less expensive.⁷ The extent to which these systems are cheaper depends partly on their requirements, such as the quality of their sensing systems, their ranges, and their payload capacities. If militaries wanted to develop extremely long-range, low-observable, and highly maneuverable uncrewed aircraft with large payloads and advanced radar, then the cost savings from removing a human pilot from the cockpit might be small. This means that militaries need to think differently about capability requirements in force design to realize cost savings. Put differently, it requires choosing to use AI to build numerous cheaper, autonomous fighters rather than exquisite, expensive ones. Policymakers might be more willing to reduce demanding requirements that have historically driven up costs partly because losing aircraft would no longer mean the loss of human life. On average, the unit price will likely decrease if militaries embrace this new approach, and economies of scale could produce a cost advantage for building larger numbers of aircraft.⁸

In addition to potentially lowering the marginal cost per unit of some aircraft, there are other ways that AI could reduce costs and, therefore, free up resources to increase force structure size. Using AI would reduce human capital requirements for skilled pilots, which are a major determinant of effectiveness and a limiting factor for scaling up the size of air forces.⁹ Without the need to continuously fly aircraft to train human pilots

⁵ For a useful introduction, see John W. R. Lepingwell, "The Laws of Combat? Lanchester Reexamined," *International Security*, Vol. 12, No. 1, Summer 1987; and Alan Washburn, *Lanchester Systems*, Naval Postgraduate School, April 2000.

⁶ Jack Watling, Oleksandr V. Danylyuk, and Nick Reynolds, *Preliminary Lessons from Ukraine's Offensive Operations, 2022–23*, Royal United Services Institute, July 18, 2024.

⁷ See, for example, Gregory C. Allen and Isaac Goldston, "Updating Augustine's Law: Fighter Aircraft Cost Growth in the Age of AI and Autonomy," Center for Strategic and International Studies, December 19, 2024.

⁸ The marginal cost of adding a highly capable AI software to a smaller and lighter aircraft could be very low, which would reinforce our postulated cost savings. This vision depends on the development of AI software that can operate within the platforms' storage, compute, and power constraints.

⁹ Rachel Cohen, "Perennial Pilot Paucity Puts Air Force in Precarious Position," *Air Force Times*, March 3, 2023.

and keep their skills sharp, aircraft maintenance requirements would likely decrease significantly. Militaries might start to treat aircraft more like missiles, in the sense that some aircraft would be flown each year to test the inventory's quality, but the aircraft would not require regular use except for employment in combat. Achieving this level of scale would require improvements to the U.S. defense industrial base, and AI's potential to generate efficiency gains in production (as outlined in Chapter 1) might help generate the kind of mass that we envision in this chapter.

At the same time, AI will likely make fighter aircraft qualitatively better in tactical engagements. In aerial combat, the side that can fuse, analyze, and act effectively on data faster than its opponent gains a significant advantage. If AI can make decisions better and faster than human pilots, it could gain a decisional advantage over crewed aircraft that translates into a combat advantage. Removing the physical constraints that come with using human pilots could also enable aircraft to operate at longer ranges without exhausting pilots and reducing their performance, as well as enable tactics that would be too dangerous or strenuous for human pilots but not for AI systems.¹⁰ For example, autonomous aircraft might become more survivable if they can react faster and make more-aggressive maneuvers than humans can to evade a missile.

How Much Quality Is Enough to Overcome Greater Quantity?

We use Lanchester equations as a framework to evaluate the consequences of changes in quantity versus quality for combat outcomes. These equations describe battles whose outcomes depend on attrition. Specifically, they are differential equations for modeling battles between two sides that last until one side loses all its forces. There are several variants of these equations, and we use the Lanchester Square Law because it best matches the character of modern warfare, especially because a core assumption of the Lanchester Square Law is that multiple units can concentrate their fire on one target.¹¹ Other key assumptions of the Lanchester Square Law equations include the following:

- There is a homogenous composition of forces on each side (e.g., fighter aircraft face other fighter aircraft, not bombers).
- A battle's outcome is based on attrition (that is, the winner is the side that grinds down and eliminates the loser's entire force; battle outcomes are not based on maneuver or operational art).
- Both sides can see and engage all the other side's forces, so the battlefield is a target-rich environment where it is possible to concentrate fire on any given target.¹²
- Both sides engage each other simultaneously, so there is no first-mover advantage; this assumption implies that neither side has such a large advantage in beyond-visual-range combat that its fighters can

¹⁰ Scharre, 2023a.

¹¹ There are two main forms of the equations—one in which each unit can attack multiple targets and receive fire from multiple directions (which Frederick Lanchester called *collective combat*) and another in which each unit can attack only one target and receive damage from one target (which Lanchester called *individual combat*). We used the collective combat model, commonly known as the *Lanchester Square Law* (Lepingwell, 1987).

Lanchester equations have historically been better suited to air combat because of its melee character; fighters have had to place themselves at risk to attack the enemy. By way of contrast, naval combat in the aviation and missile ages has depended much more on striking first effectively (see Wayne P. Hughes, *Fleet Tactics: Theory and Practice*, Naval Institute Press, 1986).

¹² For modeling simplicity, we assumed that both sides are equivalently skilled in the use of jamming and decoys such that both sides are equally able to find and engage real targets. Conceivably, AI could help one side gain an advantage in deception that changes this dynamic.

simply engage the enemy before the enemy's fighters have a chance to fight back (in other words, neither side gets to carry out a first strike).¹³

- Units retain the will to fight and keep fighting despite poor odds of survival.

AI might enable autonomous units to engage in combat in ways that more closely match the assumptions of Lanchester equations than human combat has historically been able to (Table 2.1). Analysts have often used Lanchester equations, but their simplifying assumptions mean that the equations are often poorly suited to reflect important combat dynamics. Attempts to validate Lanchester models with historical battle data have been mixed at best.¹⁴ However, a battle between autonomous robotic agents might more closely match the key assumptions behind the Lanchester equations, making the equations a useful tool for considering the implications of AI-enabled robotic warfare. Additionally, the United States has previously had large advantages in beyond-visual-range capabilities for air combat when facing such weaker regional adversaries as Iraq, violating a core Lanchester assumption. Consistent with the report's focus on long-term competition between sophisticated and adaptive peer competitors, we assume that future U.S. adversaries will also have strong beyond-visual-range capabilities, enabling both sides to engage each other simultaneously. To be clear, our argument is not that every assumption behind the Lanchester Square Law will always hold, but this framework provides a useful starting point for assessing the trade-offs between quantity and quality in robotic warfare because it is a transparent and well-established method in defense analysis. Using the assumptions also produces clear, numerical predictions, which are especially valuable for exploring the cost dynamics of this competition.

In its basic form, the Lanchester Square Law describes one force (Force A) with quantity x and an effectiveness of α . Effectiveness is measured in terms of the quantity of enemy units killed by each x per unit

TABLE 2.1

Lanchester Assumptions: Human Battle Versus Robotic Battle

Key Assumption of Lanchester Equations	Does This Assumption Apply to a Human Battle?	Would This Assumption Apply to an Autonomous Robotic Air Battle?
All forces can engage each other.	Sometimes	Sometimes
Agents are homogenous.	Sometimes	Sometimes
Agents are willing to fight despite poor odds of survival.	Rare	Yes
Raw attrition determines the battle's outcome.	Rare	Sometimes

NOTE: *Homogenous agents* in the Lanchester equations refer to symmetrical capabilities, such as fighter aircraft facing other fighter aircraft. The use of AI would likely produce more-homogenous agents at the individual level by reducing human-level variation in performance and decisionmaking.

¹³ Using AI to dramatically increase the effectiveness of beyond-visual-range combat would be one way to increase lethality, which falls outside the model's scope. However, key limitations on the variety of engagements are more likely to be sensors' physical limits (e.g., their line of sight to the horizon) and the rules of engagement (e.g., ensuring that a fighter is about to engage an enemy aircraft and not a friendly or civilian aircraft).

¹⁴ For example, Engel found a reasonable fit for the Battle of Iwo Jima if one included American reinforcements, but other scholars have failed to find good fits for Lanchester equations to the Battle of the Ardennes in 1940 and the Battle of Kursk in 1943 (see J. H. Engel, "A Verification of Lanchester's Law," *Journal of the Operations Research Society of America*, Vol. 2, No. 2, May 1954; see also Jerome Bracken, "Lanchester Models of the Ardennes Campaign," *Naval Research Logistics*, Vol. 42, No. 4, June 1995; and Turker Turkes, *Fitting Lanchester and Other Equations to the Battle of Kursk Data*, thesis, Naval Postgraduate School, 2000).

time t . Force A is in combat with a second force (Force B) with quantity y and effectiveness β . The derivatives of x and y with respect to t are as follows:¹⁵

$$\dot{x} = -\beta y$$

$$\dot{y} = -\alpha x.$$

To summarize, in the simplified world of Lanchester combat, there are **only two ways to improve the combat power** of one's force: **increase lethality** or **increase quantity**.

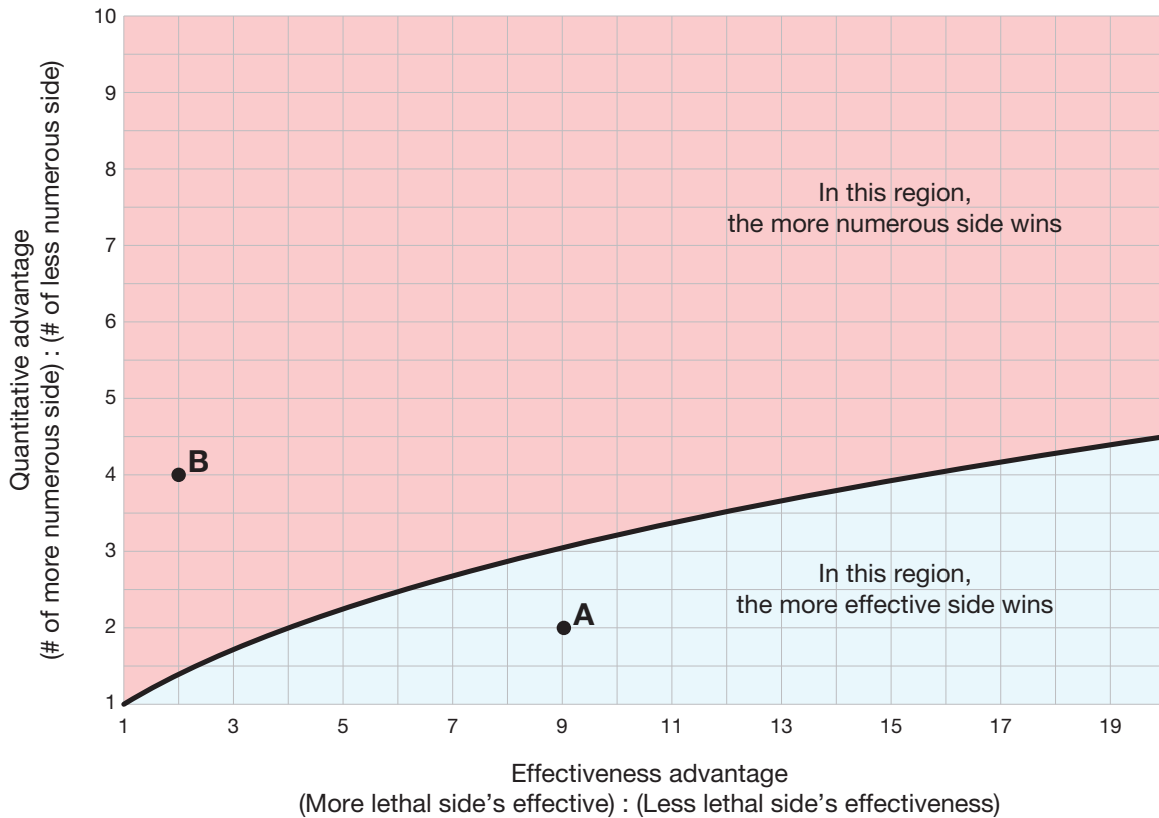
- **Increase lethality.** AI could improve the lethality of aircraft by helping aircraft fuse data and act to exploit opportunities faster and more effectively than human pilots can. However, there are likely limits on the magnitude of potential lethality improvements. For example, achieving increases of multiple orders of magnitude would require breakthroughs in engineering to increase the number of potential kills that finite platforms can produce, which makes it less plausible that AI by itself will enable a revolution in lethality. Additionally, if such qualitative improvements as improved response times or maneuverability increase one side's survivability, that will decrease the other side's relative lethality. However, the nature of Lanchester combat means that chasing lethality offers less leverage to a force planner than pursuing quantity.
- **Increase quantity.** AI might enable levels of good-enough autonomy that, when combined with procurement of aircraft in sufficient numbers, can prevail even over opponents that are more lethal on a per-unit basis. For example, the USAF would not need to field a collaborative combat aircraft (CCA) with AI that is as lethal as a crewed Chinese fighter to realize an advantage. If the USAF could field twice as many CCA than Chinese fighters in a battle, then the CCA could win, as long as their lethality is not less than one-quarter of that of the Chinese fighters.

If the Lanchester Square Law accurately captures key dynamics for air-to-air engagements enabled by new AI breakthroughs, then quality will face an uphill battle against quantity. One of these equations' benefits is that the equations clearly identify the frontier at which advances in quality could, in fact, overcome disadvantages in quantity. We can visualize how much of a quality advantage would be necessary to overcome a given numerical disadvantage (obviously, if one side has both a quality and quantity advantage, then that side wins). Figure 2.1 shows a graph of the Lanchester Square Law. In the blue region below the curve, the more effective side (in terms of lethality) wins. For example, a side with a 9:1 advantage in lethality can win if it is outnumbered 2:1, as shown by point A. In the red region above the curve, the more numerous side wins. For example, a side with a 4:1 quantitative advantage can win even if its adversary has a 2:1 advantage in lethality, as shown by point B.

Historical examples can provide a sense of what a realistic upper bound on lethality advantages might be (absent a radical breakthrough in physics that could enable completely new modes of combat). Building on Figure 2.1's model, Figure 2.2 adds a historical example from air-to-air combat during World War II. Late in the war, Germany deployed the Messerschmitt Me 262, the world's first jet fighter (U.S. aircraft, such as the Mustang P-51D fighter, still used piston engines at the time). The Luftwaffe generally staffed Me 262 units with extremely experienced German aces. A battle between the Me 262 and the P-51D is a useful historical illustration of a breakthrough technology giving one side a significant lethality advantage. Because of these advantages in superior technology and pilot skill, historical data on air-to-air engagements

¹⁵ Washburn, 2000.

FIGURE 2.1
Quality Versus Quantity Break-Even Points (Using the Lanchester Square Law)

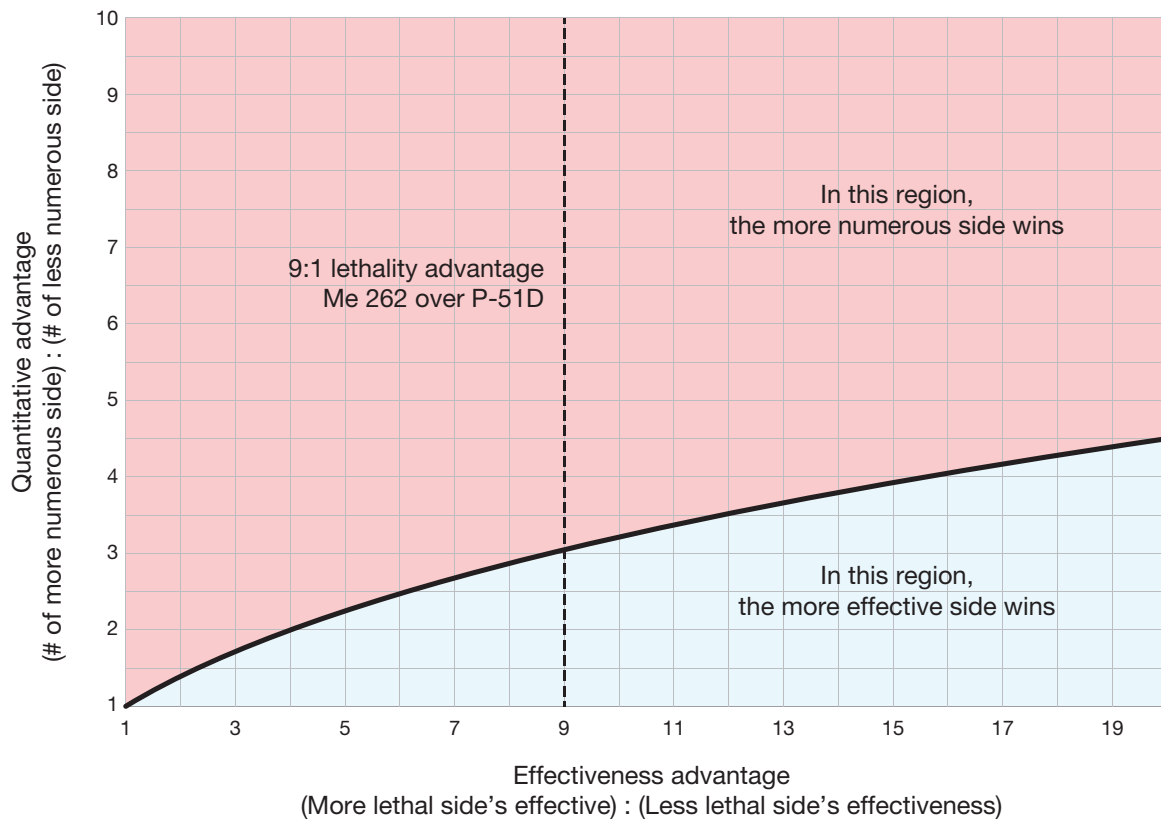


between the Me 262 and the P-51D suggest that the Me 262 likely had around a 9:1 lethality advantage over the P-51D.¹⁶ Using the Lanchester Square Law, that suggests that the Me 262 fighter could overcome being

¹⁶ To estimate the Me 262's lethality advantage, we drew on two days of intense aerial combat in 1945 that represent the peak of the Me 262's use in the war (John Foreman and S. E. Harvey, *The Messerschmitt Combat Diary Me.262*, Crecy Pub, 2nd ed., 1995). On March 18, 1945, the United States generated 600 fighter sorties compared with 37 German sorties of Me 262s. Germany achieved 21 kills against U.S. fighters, and the United States killed six Me 262s. On April 10, the United States generated 815 fighter sorties compared with 55 German sorties of Me 262s. Germany achieved four kills against U.S. fighters, while the Germans lost 33 Me 262s. Combining the statistics from these days, the United States had 1,415 fighter sorties, Germany had 92 fighter sorties, the United States had 39 fighter kills, and Germany had 25 fighter kills. Determining the lethality advantage for the Lanchester Square Law requires accounting for the number of fighters involved in engagements. Looking solely at loss-exchange ratios conflates advantages derived from quantitative advantages with advantages from qualitative advantages in lethality. When adjusting for the sortie ratio, we found that the Me 262 had a lethality advantage of 9.86 ($1,415/92 \times 25/39 = 9.86$). In our analysis, we rounded down from 9.86 to 9 because of the illustrative simplicity of using a perfect square and because the difference between the two numbers is too small to change our conclusions. These statistics are intended to provide an illustrative upper bound, but they are imperfect, and different approaches to counting could lead to slightly different estimates. These were not truly "pure" fighter-versus-fighter duels between the Me 262 and the P-51D. The Me 262 was also targeting U.S. bombers, but the imperative to hunt bombers is what forced German fighters to engage U.S. fighters in the first place. Additionally, on March 18, only ten of the 21 U.S. fighter losses were P-51s. If the other fighter types are excluded, the Me 262's relative lethality against the P-51 would fall to 5.52. By contrast, on April 10, U.S. fighters destroyed three Me 262s in a strafing run while the Me 262s were still on the ground. Excluding those losses would increase the Me 262's relative lethality to 10.68. Making both adjustments would lead to an Me 262 lethality advantage of around 6:1, which would fall even more if the data were available to reduce the U.S. sortie rate on March 18 to include only P-51s and no other fighter types. The estimate of a 9:1 advantage is therefore well grounded empirically as a strong upper bound on what the Me 262

FIGURE 2.2

Quality Versus Quantity Break-Even Points with Historical Example



outnumbered by 3:1. The Me 262 example is a reasonable upper bound on the maximum advantage that superior equipment might provide against a skilled adversary.

We are using this historical example of a previous technological leap in air-to-air combat to help bound the scope of the problem and ground the analysis; it is not meant as a prediction of AI's effects. If advantages in relative lethality are significantly greater than 9:1, then the enemy is unlikely to be a peer competitor, which was the case in Israel's lopsided air-to-air combat victories against much-weaker Arab militaries in the past.¹⁷ If 9:1 provides a reasonable upper bound on the advantages that next-generation technologies provide against a peer adversary, then, at best, lethality improvements might offer the ability to fight outnumbered 3:1. Therefore, if AI can enable more than a threefold increase in the quantity of systems that are at least one-ninth as lethal as what the adversary fields, then the side with the larger force would prevail.

This analysis supports the hypothesis that AI could give quantity an edge over quality. Fielding large numbers of systems at a reasonable quality is more valuable than chasing extreme lethality improvements. This implies that an investment strategy to create a mass of good-enough AI-piloted fighters would outperform an investment strategy to use AI to enhance the lethality of exquisite fighters; a force of 401 drones

might have had over the P-51. The overall results of the analysis are not sensitive to modest adjustments to this upper bound. For excellent historical analysis on trends in fighters' qualitative advantages that supports the assessment that a 9:1 advantage was still an upper bound in subsequent decades, see John Stillion, *Trends in Air-to-Air Combat: Implications for Future Air Superiority*, Center for Strategic and Budgetary Assessments, 2015.

¹⁷ Kenneth M. Pollack, *Arabs at War: Military Effectiveness, 1948–1991*, University of Nebraska Press, 2002.

would defeat a force of 200 drones, even if the 200 drones were four times as lethal as the 401 drones. Put differently, one should choose a force of 950 autonomous P-51s over 300 autonomous Me 262s, so long as the autonomous P-51s are at least one-ninth as lethal as the autonomous Me 262s.

What Are Potential Limiting Factors on Embracing Mass?

Limiting Factor 1: Cost

If the Lanchester Square Law suggests that quantity has such a strong advantage over quality, why have more militaries *not* focused their force structures more squarely on mass? One potential reason is that the assumptions of the Lanchester Square Law are difficult to map onto the dynamics of human combat, causing the Lanchester Square Law to overstate the advantages of quantity. But if the assumptions more closely resemble robotic warfare, the relative advantages of quantity might increase in the future.

Another potential reason is that generating enough mass is expensive. If a poorer country is generally the one trying to rely on sheer numbers to offset the qualitative advantages of an opponent, then it must generate enough force to counterbalance its lethality disadvantage. Relying on sheer numbers can be quite expensive, especially for such systems as fighter aircraft. This is especially true because, as our break-even analysis highlights, a mass strategy still requires fielding units that have a sufficient baseline of lethality, which drives up cost. The logistical burdens of supporting a much larger force structure could also increase costs, although further analysis is necessary to compare these costs with those of supporting exquisite aircraft.

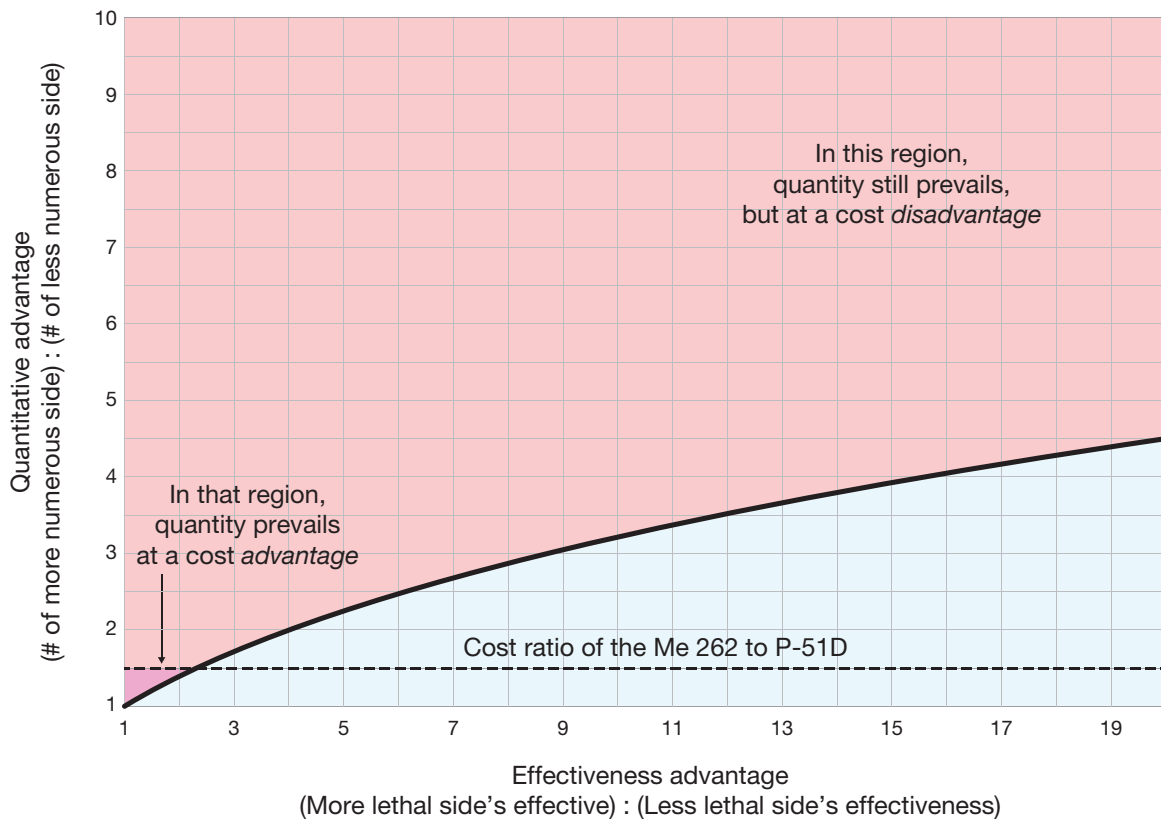
This cost consideration suggests that quantity might have a combat advantage at the tactical level, but generating a much larger force structure might be cost-ineffective and financially unsustainable at the strategic level. Even if the unit price is cheaper, the cost of the total force structure might be more expensive. In a long-term competition in which neither side can afford to dramatically outspend the other, investing in mass might be impractical.

We can see how relying on quantity can cost more than relying on quality by looking at the historical example introduced earlier. Although the P-51D could defeat the Me 262 by deploying in greater quantities, this strategy of relying on quantity to overcome quality might have cost the United States more than Germany. Using cost-estimating relationships that are based on the aircraft characteristics, such as weight and speed, we estimated that an aircraft as capable as the Me 262 costs about 1.5 times more than an aircraft as capable as a P-51D.¹⁸ Figure 2.3 adds a horizontal line (reflecting the cost ratio for an Me 262 relative to a P-51D) to the earlier break-even analysis from the previous figures. In the red region above the curve, the quantitatively larger force of P-51Ds still prevails over the smaller force of Me 262s, but the victorious P-51D force costs more than the Me 262 force. That is, the United States could win the battle, but it would pay disproportionately more for the victory. It is only in the relatively small purple region that is above the curve but below the dotted line that the United States would both win and do so at a cost advantage over Germany. Given the estimated 9:1 lethality advantage of the Me 262 over the P-51D, this means that the United States would have to spend more than Germany to prevail in a battle between P-51Ds and Me 262s. For the United States in World War II, these kinds of cost inefficiencies were acceptable because the United States could out-produce Germany. In a competition between two countries with equivalently strong defense industrial base capacities, that might not be the case.

However, AI might change the dynamics of this cost competition. As discussed previously, AI could make it possible to field much-cheaper platforms without a correspondingly steep reduction in combat capability. Figure 2.4 visualizes the same break-even analysis as in the previous figures, but it uses the cost

¹⁸ See the appendix for an explanation of cost-estimating relationships and how we applied them here.

FIGURE 2.3
Cost Calculations and the Affordability of Quantitative Advantages

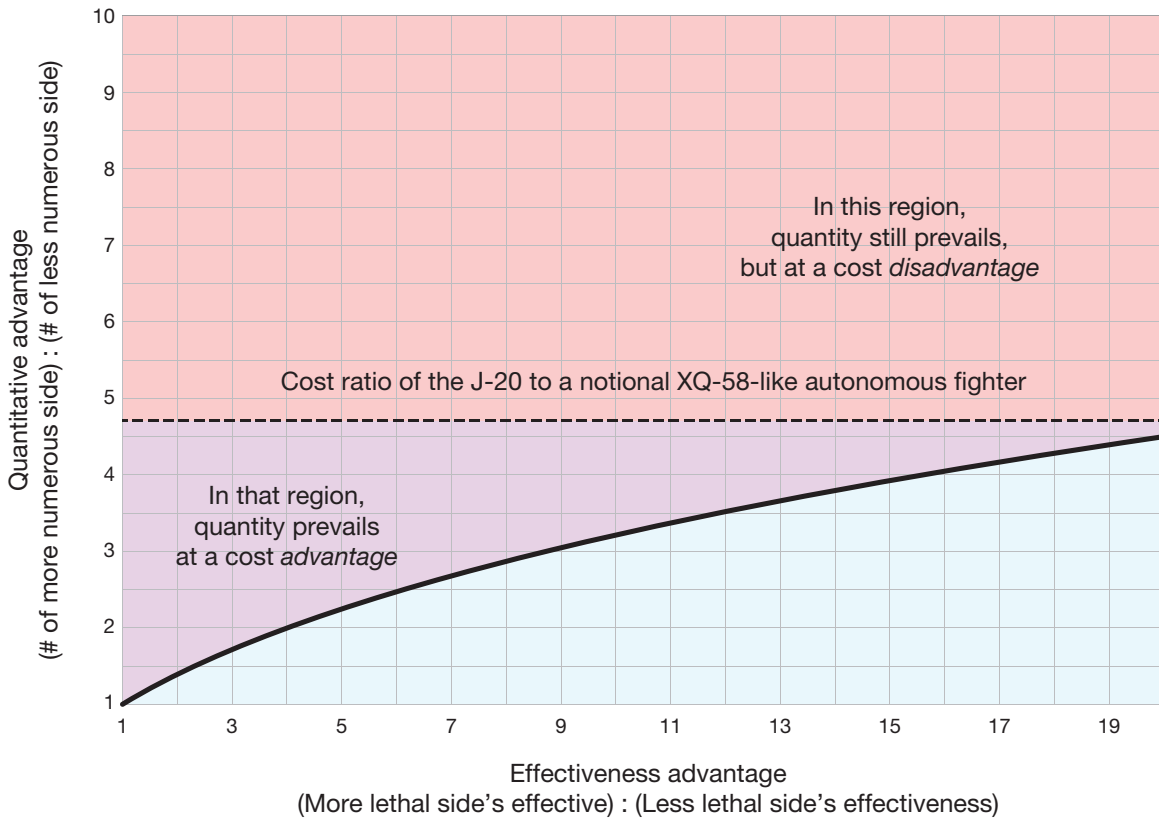


ratio from a forward-looking case that compares exquisite and attritable fighters rather than a historical case. Figure 2.4 plots the estimated cost ratio of a J-20, a crewed fifth-generation fighter from China, compared with a notional autonomous robotic fighter that has characteristics comparable with the XQ-58 Valkyrie, which is an experimental uncrewed combat aerial vehicle (UCAV) from the United States. Using a basic cost-estimating relationship, we estimate that the cost of one J-20 might be around 4.71 times as much as a robotic fighter of similar size to the XQ-58.¹⁹ Figure 2.4 illustrates that the hypothesized cost reduction significantly expands the purple region in which robotic mass prevails at a cost advantage over exquisite fighters. AI's most-dramatic impact, therefore, might be enabling cost-effective mass in a way that was previously not possible.

¹⁹ The parameters and their coefficients are as follows: a dummy variable for a remotely piloted vehicle (0.299), the log of empty weight (0.651), a dummy variable for stealth (0.747), the log of the production rate (−0.345), the year of the first delivery (0.006), an indicator for the mission type (0.339 for fighters), and an intercept (−13.123). The dependent variable is the log of the average unit flyaway costs of the first 100 aircraft. For our purposes, we were interested in comparing the relative costs of two different types of aircraft, so the exact form the costs take in absolute terms is less important, as are such variables as the year of the first delivery—as long as their values are held constant across the two aircraft of interest, which would allow us to focus the comparison on differences in capability attributes (see Thomas Light, Fred Timson, and Obaid Younossi, *Parametric Flyaway Cost Estimating Relationships for Manned and Remotely Piloted Military Aircraft*, RAND Corporation, RR-1618/1-AF, 2024, p. 11). The estimate of the empty weight for the J-20 (42,750 pounds) came from Janes, and the estimate of the empty weight for the XQ-58 (2,500 pounds) came from a fact sheet produced by the aircraft's manufacturer, Kratos (Janes, "J-20," webpage, last updated June 19, 2025; Kratos, "Valkyrie XQ-58A," fact sheet, undated).

FIGURE 2.4

How Autonomy Might Change the Cost of Relying on Quantity



More-detailed estimates of the magnitude of potential cost advantages in a mature AI regime (in which both sides simultaneously field new AI-enabled aircraft) are needed and would require further analysis. Figure 2.4 compares an attritable robotic fighter with a contemporary exquisite aircraft, but the costs of uncrewed exquisite aircraft could be either greater (if additional requirements were added, such as fielding uncrewed sixth-generation fighters) or lesser (if militaries fielded uncrewed variants of fifth-generation fighters that had similar attributes but were designed to reduce costs). However, as we noted in the section titled “How AI Can Help Both Quantity and Quality in Air Combat,” there are limits to how much cheaper exquisite aircraft can be. This is because the nature of advanced technical capabilities, such as stealth and advanced radar, means that the presence of a human in the cockpit has a smaller impact relative to these other attributes on the aircraft’s overall cost.

Limiting Factor 2: Geography

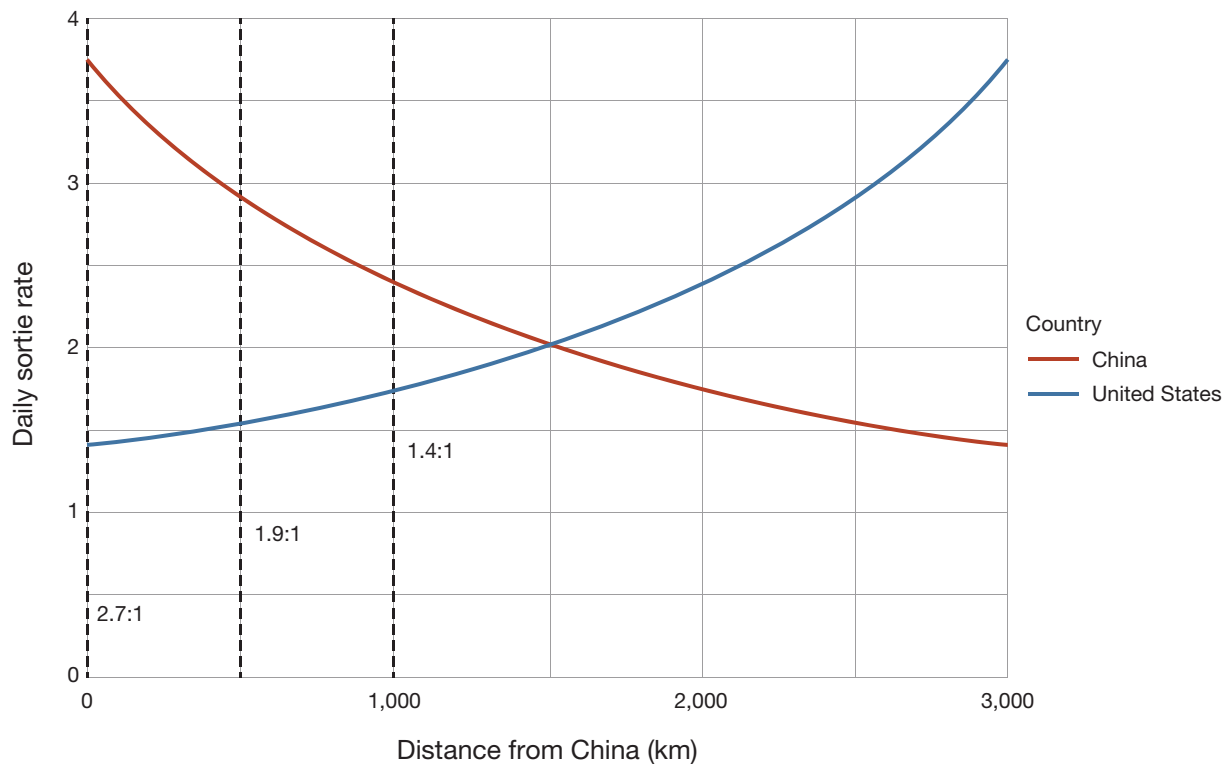
Geography is another variable that could qualify our findings. Distance is a factor in each side’s ability to bring mass to bear in each battle. The farther one side’s forces must fly from an air base to a battle, the fewer sorties a force of a given size will be able to generate.²⁰ A farther distance can also require that one side field more-capable and more-expensive aircraft that have greater range. The closer one side’s air bases are to

²⁰ John Stillion and David T. Orletsky, *Airbase Vulnerability to Conventional Cruise-Missile and Ballistic-Missile Attacks: Technology, Scenarios, and U.S. Air Force Responses*, RAND Corporation, MR-1028-AF, 1999, pp. 81–84.

a battle, the more sorties it can generate and the more viable it is to build cheaper, shorter-range fighters. Asymmetries in geographic contexts can make the more-distant side build larger, more-expensive aircraft and force it to build more of them to reach quantitative parity with the side that is geographically closer to the battlefield.

Drawing on the U.S.-China fighter context we presented for Figure 2.4, Figure 2.5 applies a simple sortie rate model to provide a basic illustration of how geography imposes its own constraints on mass.²¹ The closer that air-to-air engagements are to China, the farther they occur from U.S. air bases in the region, such as those in Japan or Guam. In a case in which the engagement occurs right off China's shores, China has a major advantage in generating sorties because its aircraft do not need to fly for very long, so it can keep larger numbers of aircraft on station for longer periods. All else being equal, each side operating the same number of aircraft from different distances would give China a 2.7:1 force advantage over U.S. fighters along its coast, although the size of this advantage would diminish as air-to-air engagements took place farther from China. If the United States wanted to achieve quantitative parity in air-to-air engagements close to China, it would need to use a larger force structure than China (or else it would need to leverage the element of surprise). If

FIGURE 2.5
The Relationship Between Distance and Sortie Generation



NOTE: The ratios reflect China's sortie advantage at a given distance from its borders.

²¹ To generate the curves in Figure 2.5, we varied the distance to identify the impact on sortie rates by directly replicating Stillion and Orletsky's model (Stillion and Orletsky, 1999). The assumptions in this model include that the aircraft flies at 500 knots for its cruising speed, that the flight time is double the distance divided by the aircraft's speed (to capture the round trip of a sortie), that the aircraft's turnaround time for such tasks as landing and refueling is three hours, and that the maintenance time on the ground is 3.4 hours plus the number of hours the aircraft flew multiplied by 0.68. The daily sortie rate is then 24 hours divided by the sum of the aircraft's flight time and ground time, with the ground time including both the turnaround time and maintenance time.

both sides developed robotic air superiority aircraft, all else being equal, the U.S. aircraft would need to be more expensive to give them greater range if they needed to operate from distant bases, such as Guam (or the United States would need to use transport aircraft to get shorter-range aircraft closer to the fight).

In the context of a war over Taiwan, if the United States were to follow its legacy approach to power projection, it would operate from a small number of air bases that would be farther from the battlefield than China's air bases. If the United States maintained this approach, then the geographic asymmetry would prevent the United States from achieving the full potential of autonomous fighters' cost advantages that we discussed previously. Smaller autonomous fighters could still offer advantages, but not as much as postulated in the XQ-58 example. If the United States does not change its approach to power projection, then there is reason to moderate expectations that autonomy will create a large U.S. cost advantage over China.

If, however, the United States were to explore a new approach to power projection that relied on large quantities of runway-independent UCAVs, then the United States might be able to realize the aforementioned advantages by basing these UCAVs in multiple distributed sites across Taiwan, southwest Japan, and the northern Philippines.²² Adopting this concept would help mitigate the impact of the missile threat to U.S. and allied air bases in the region by detaching some airpower from fixed operating locations that Chinese cruise and ballistic missiles could easily target.²³ Massed autonomous UCAVs could help accomplish combat missions, and their quantity could overcome their reduced effectiveness compared with Chinese fighters.

Limiting Factor 3: Generalizability to Other Domains

Although our analysis focuses on air-to-air engagements, quantity might have growing advantages in other domains as well. The naval domain is a clear area for future extensions of this analysis. Advances in China's space and long-range strike capabilities are posing a major challenge to large U.S. surface combatants, reinforcing long-standing calls for the United States to revisit its mix of large surface ships versus smaller, uncrewed, and undersea capabilities.²⁴ Extending the Lanchester equations' assumptions to naval warfare requires some modification because of the importance in the missile age of timing salvo pulses.²⁵ There are large advantages of firing first in naval warfare that the Lanchester equations do not fully capture. Additionally, there are other aspects of naval warfare that require different kinds of analysis, such as how having fewer or no crew on board changes the ability to conduct maintenance and battle damage control on a platform. We still expect robotic mass to create new opportunities in naval warfare, but these are reasons to be cautious about extrapolating conclusions from our air-to-air engagement examples to naval operations without additional analysis.

Another area for further research is how our hypothesis—about the impact of AI on the trade-offs between quality and quantity—applies to ground forces. Although Lanchester equations have been applied to ground force combat, validating them with historical ground combat data has been mixed.²⁶ Furthermore, identifying likely cost curves and force structure mixes for ground forces is more complex than for the fighter

²² See, for example, Thomas Hamilton and David A. Ochmanek, *Operating Low-Cost, Reusable Unmanned Aerial Vehicles in Contested Environments: Preliminary Evaluation of Operational Concepts*, RAND Corporation, RR-4407-AF, 2020.

²³ The dispersed launch and recovery elements that operate the runway-independent UCAVs could employ the deception methods we discuss in Chapter 3 to improve their survivability.

²⁴ Audrey Decker, "Chinese Satellites Are Breaking the US 'Monopoly' on Long-Range Targeting," *Defense One*, May 2, 2024; Scott Savitz and Amanda Perez, *Could the U.S. Navy Fleet of the Mid-21st Century Include Large Uncrewed Vehicles?* RAND Corporation, PE-A2854-1, January 2025.

²⁵ Hughes, 1986, pp. 37–38, 271–272.

²⁶ Engel, 1954; Bracken, 1995; Turkes, 2000.

cost-estimating relationships examined in this chapter. Future extensions of the quantity-quality analysis to the ground domain would therefore likely require a different framework than what we used in this chapter.

Conclusion

We hypothesize that AI could give quantity a relative advantage over quality. Fielding large numbers of good-enough robotic fighters at scale could improve the cost-effectiveness of relying on mass over quality. Our analysis focuses on the context of air warfare, but this trend could apply to other domains as well.

The growing advantages of quantity over quality represent a reversion, in some ways, to long-term historical trends. For most of history, militaries have viewed mass as a key factor in giving one side an edge on the battlefield.²⁷ Militaries could sometimes offset quantitative disadvantages if they could leverage technological innovations faster or better than their opponents. British colonial troops armed with the Maxim gun in colonial conflicts serve as an infamous example.²⁸ These types of qualitative advantages were generally short-lived and did not change the larger preference for quantity over quality once a technology had diffused. Toward the end of the Cold War, however, the United States pioneered advances in stealth and precision-strike networks that created overwhelming and long-lasting qualitative advantages. Although such exquisite capabilities as stealth aircraft significantly drive up costs, their performance advantages have made it more cost-effective to rely on fewer but better platforms and weapons for the past 50 years. Improvements in robotics and autonomy are starting to erode these advantages. As lower-end uncrewed systems become more capable without a correspondingly large rise in cost, relying on a small number of qualitatively superior platforms will become less cost-effective. Although exquisite platforms certainly have a role in future battlefields, the optimal mix between quality and quantity could increasingly favor quantity.

In a competitive environment, the quantity-quality competition will continue to evolve over time. If China can also embrace an AI-enabled quantity approach, then its geographic and industrial base advantages could provide it with a powerful edge in an AI-enabled quantity-based competition. In such a world, the United States might need to pursue strategies to break free of Lanchester dynamics to regain an advantage (such as by breaking the assumption of simultaneous engagements and creating a first-look, first-shot, first-kill advantage).

²⁷ Horowitz, 2024.

²⁸ Scharre, 2018, p. 38.

Hider Versus Finder

How will AI affect the relative advantage between hiders and finders in the scouting competition?¹ There is a long-standing belief among certain analysts and policymakers that technological advances will eventually create a “transparent battlefield.”² Some now believe that AI could make this vision a reality by fusing and analyzing intelligence from proliferated sensors.³ The head of the U.S. Army Futures Command predicted that, by 2040, the “ability to hide, which is fundamental to how we fight, [will be] impossible.”⁴ Advances in AI certainly have the potential to make reconnaissance more effective and to close kill chains faster, but visions of AI providing finders with decisive advantages rest on assumptions about an ideal environment for the finder and overlook countermeasures available to the hider.

Hiders will also benefit from AI, and militaries will not stand still as their adversaries get significantly better at reconnaissance. Some countermeasures are low-tech, such as attacking sensors and controlling movements or electronic emissions. Other countermeasures are high-tech, such as fielding autonomous robotic decoys or developing an AI fog of war machine, which is an AI tool that can help plan and orchestrate elaborate deception campaigns designed to maximize the finder’s confusion and uncertainty.⁵ Although AI’s potential to enable finding has received the lion’s share of attention, AI could grant hiders a cost-effective option to offset improvements to an enemy’s sensing under certain conditions. The United States, therefore, cannot count on AI-enhanced situational awareness providing a transparent battlefield because of these countermeasures. The United States also needs to invest in its own automated military deception to similarly maximize its opponents’ confusion and uncertainty.

The hider-finder competition will likely remain especially contested, and neither side will gain an unequivocal edge.⁶ Three variables will influence whether the hider or finder gains an advantage in specific settings. First, the hider-finder competition interacts with the quality-quantity competition. If a finder can field far more sensors than a defender can field decoys, the finder has better chances of reducing uncertainty from the defender’s deception campaign. Second, the ability to create uncertainty depends on the kind of

¹ In this chapter, we primarily focus on the scouting competition at the tactical level and touch on some implications for the operational level. The strategic level is beyond this chapter’s scope because it involves different considerations.

² Other analysts have also expressed skepticism about a transparent battlefield. See Ivan Oelrich, Paul van Hooft, and Stephen Biddle, “Anti-Satellite Warfare, Proliferated Satellites, and the Future of Space-Based Military Surveillance,” *Journal of Strategic Studies*, Vol. 47, No. 6–7, 2024.

³ See, for example, “The Added Dangers of Fighting in Ukraine When Everything Is Visible: There Is No More Fog of War,” *The Economist*, February 6, 2025; and David Barno and Nora Bensahel, “The Other Big Lessons That the U.S. Army Should Learn from Ukraine,” *War on the Rocks*, June 27, 2022.

⁴ Stew Magnuson, “Future Command’s Rainey Eyes the Army of 2040,” *National Defense*, October 6, 2023.

⁵ Edward Geist, *Deterrence Under Uncertainty: Artificial Intelligence and Nuclear Warfare*, Oxford University Press, 2023.

⁶ Others have also used this hider-finder construct. See, for example, Krepinevich, 2015, pp. 65–66; and Antonio Calcara, Andrea Gilli, Mauro Gilli, Raffaele Marchetti, and Ivan Zaccagnini, “Why Drones Have Not Revolutionized War: The Enduring Hider-Finder Competition in Air Warfare,” *International Security*, Vol. 46, No. 4, Spring 2022.

information in question. It might be very difficult to prevent a finder from having a general sense of where large formations of forces are operating, but it might be much easier for a hider to create uncertainty about the exact locations of smaller high-value targets. Third, there is variation across different domains and physical environments.⁷ Militaries can leverage AI in various ways to increase the amount of noise, confusion, and clutter in all domains, but not all domains provide equal opportunities to do so. Wider and more-cluttered search areas lend themselves to hiding, and smaller and emptier areas lend themselves to finding. The ground domain and undersea domain provide the strongest opportunities for hiding and deception, followed by the air and sea surface, followed by space as the weakest domain for deception.

How AI Can Help Finders

The typical vision of how AI will help finders is through information-fusion systems, which combine large amounts of data from various sources to build up an increasingly detailed picture of the state of the world. AI can help militaries field more sensors and collect more data. For example, hundreds or thousands of AI-enabled drones could work together to form resilient “targeting meshes” that provide sensing in contested areas.⁸ AI can also help automate data-processing. This could include automated target recognition for drones at the tactical edge. It could also mean replacing teams of human intelligence analysts with automated systems that can operate at a speed and scale that is impractical for humans.⁹ For example, a new generation of small satellites collects terabytes of imagery data every day, and making sense of this much data within an operationally relevant time frame is only possible with automation.¹⁰ Taken together, AI could help militaries field larger and more-capable sensing networks, as well as analyze the data that flow through those networks more quickly and effectively.

From the finder’s perspective, these advances in AI-enabled sensing can make kill chains faster and more lethal. Figure 3.1 illustrates the links of the kill chain. In 1996, USAF Chief of Staff General Ronald Fogleman predicted that, “in the first quarter of the twenty-first century, it will be possible to find, fix or track, and target anything that moves on the surface of the earth.” *Find, fix or track*, and *target*, with the addition of *engage* and *assess*, form the links of the kill chain.¹¹ The usual intuition is that AI will strengthen kill chains by helping connect the find and fix/track link, as well as assisting in the assess link. Autonomous weapon systems and AI-enhanced decision-support systems could also help connect the fix/track, target, and engage links, such as by helping manage battle networks that dynamically cue strike assets to engage time-sensitive targets once sensors locate them.

Although AI has clear benefits for reconnaissance, there is a risk of overstating the extent of the finder’s potential advantages. Some analysts have pointed to Ukraine as a preview of how using ubiquitous sen-

⁷ As we discuss later in this chapter, the relative difficulty and efficacy of deception and counter-deception will depend strongly on the combat domain. In the air and space domains, real targets look very different from naturally occurring confounders (large metal objects do not fly through the air without human intervention), but, in the ground domain, the surface of the earth provides a source of confusion and clutter.

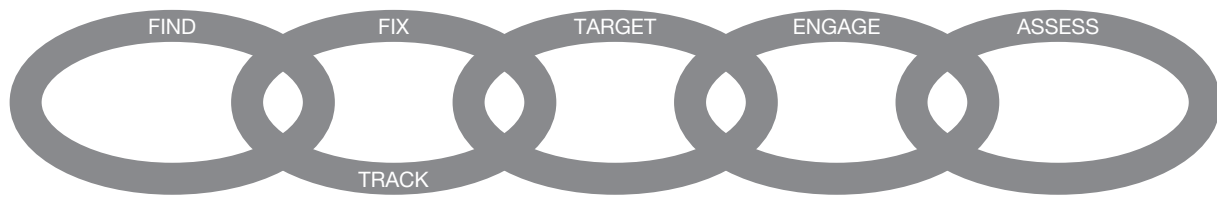
⁸ Hamilton and Ochmanek, 2020.

⁹ Chang, 2021, p. 36.

¹⁰ Saadia M. Pekkanen, Setsuko Aoki, and John Mittleman, “Small Satellites, Big Data: Uncovering the Invisible in Maritime Security,” *International Security*, Vol. 47, No. 2, Fall 2022.

¹¹ Geist, 2023, p. 85.

FIGURE 3.1
The Five Links of the Kill Chain



SOURCE: Adapted from Geist, 2023, p. 85.

sors and automated data analysis will make battlefields transparent and lift the fog of war.¹² But identifying a more realistic upper bound on that improvement is important for understanding how the measure-countermeasure race will evolve. Using AI to process and analyze information is straightforward if the world is not too complicated and the data that the sensing network collects are comprehensive and consistent. In practice, this is rare. The world is constantly changing and the information available is frequently misleading, often because of random noise and sometimes because of willful deceit. The flawed information that is available reflects only a small subset of the overall state of the world. An AI information-fusion system must (1) consider all the possible states of the world that might be consistent with the imperfect information it has available and (2) determine which of those states is most plausible. The system must make that determination with the knowledge that it certainly has incomplete information and that it might also have inaccurate information. Therefore, information fusion is a form of reasoning under uncertainty.

Information fusion has inherent logical and computational constraints because reasoning under uncertainty is hard for both machines and people.¹³ AI researchers have spent decades attempting to find a way to make computers reason well under uncertainty, because doing so is necessary for behaving intelligently. In the process, researchers developed an increasingly sophisticated theoretical understanding of reasoning under uncertainty. Several decades ago, this research revealed that a well-studied problem in theoretical computer science that is equivalent to information fusion—inference in Bayesian belief networks—is provably *nondeterministic polynomial-time hard* (NP-hard).¹⁴ This means that no computationally tractable approximation algorithm can guarantee whether it will find a solution of the true value within a certain time interval. Additionally, it turns out that there is not one right way to represent and reason about uncertainty

¹² See “The Added Dangers of Fighting in Ukraine When Everything Is Visible: There Is No More Fog of War,” 2025. For an assessment of these claims, see Mark Hvizda, Bryan Frederick, Alisa Laufer, Alexandra T. Evans, Kristen Gunness, and David A. Ochmanek, *Dispersed, Disguised, and Degradable: The Implications of the Fighting in Ukraine for Future U.S.-Involved Conflicts*, RAND Corporation, RR-A3141-2, 2025.

¹³ Edward Geist, “Why Reasoning Under Uncertainty Is Hard for Both Machines and People—and an Approach to Address the Problem,” in Aaron B. Frank and Elizabeth M. Bartels, eds., *Adaptive Engagement for Undergoverned Spaces: Concepts, Challenges, and Prospects for New Approaches*, RAND Corporation, RR-A1275-1, 2022.

¹⁴ Computer scientists sometimes use *NP-hard* as a shorthand for problems that are computationally intractable in the general case. *NP*, or nondeterministic polynomial time, is a complexity class containing decision problems (problems with definitive “yes” or “no” answers) that can be computed in polynomial time on a nondeterministic Turing machine, which is a hypothetical type of computer that could evaluate multiple branches simultaneously. This class of decision problem is of particular interest to theoretical computer scientists because the answers to such questions can be checked “quickly” using a deterministic Turing machine. And problems in which P (or polynomial time) = NP might be the hardest decision problems that are computationally tractable using classical computers. *NP-hard* problems are those that can be reduced in polynomial time using a deterministic Turing machine to all problems in *NP*; therefore, a fast solution to any *NP-hard* problem is also a shortcut to solving every problem in *NP*. Some problems in *NP* are themselves also *NP-hard*; these problems are called *NP-complete*. Most *NP-hard* problems, however, are in complexity classes that include problems harder than those in *NP*.

because there is more than one way to be uncertain. This means that employing a bigger computer, more sensors, or even some form of superintelligent AI does not necessarily overcome obstacles to information fusion.¹⁵ The targeting process does not require a guaranteed solution to uncertainty, only sufficiently accurate probabilistic assessments, but the challenge of reasoning under uncertainty places an upper limit on the quality of those AI-enabled assessments in real-world settings.

Although AI information-fusion systems will certainly be capable and useful, this means that even highly advanced AI faces limitations in its ability to make sense of complex environments. AI cannot analyze information it does not have. If a physical sensor is not in the right place or does not survive long enough to collect intelligence on a target, that is not a problem that AI can solve through better reasoning. And, for the information that sensors *do* observe, even advanced AI might struggle to generate high-confidence probabilistic assessments of what are real targets against complex background environments with a lot of noise and clutter. If the finder does not need much confidence and is willing to attack everything it finds, that might not be a problem. In cases in which the attacker wants some meaningful level of confidence that it is attacking what it intends to attack, the key considerations for the hider-finder competition then become (1) the conditions under which this task is most difficult for the finder and (2) how a hider can best aggravate this challenge.

How AI Can Help Hiders

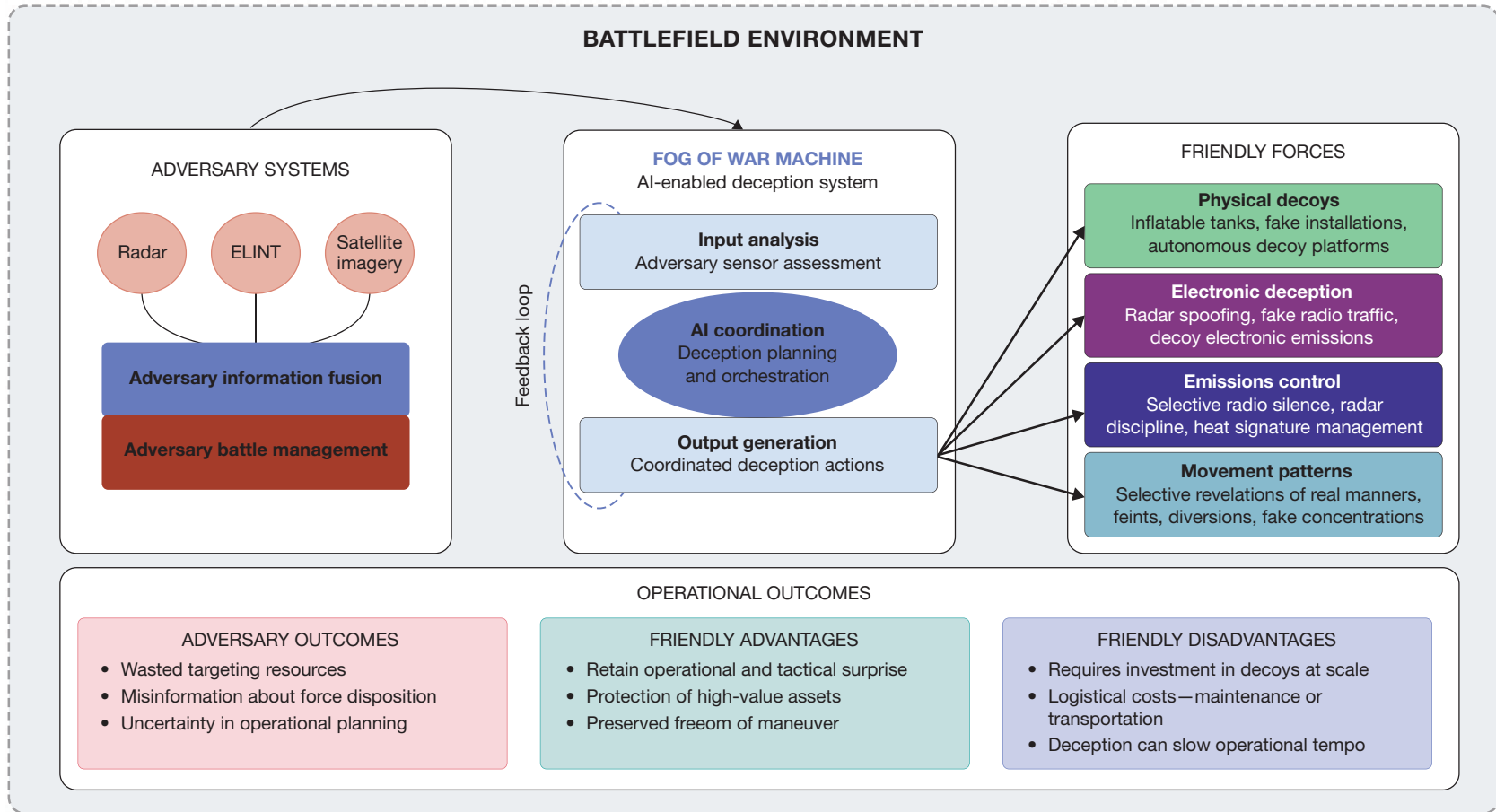
The key task for a hider is to complicate a finder's operational picture by creating uncertainty. This does not necessarily require literally hiding targets in the sense that they are not visible, although concealment is sometimes an option. We instead focus on deception as a method to counter improvements in finding by aggravating the challenge of reasoning under uncertainty. This means that the hider's goal is not necessarily to conceal any one target but to create a needle-in-the-haystack problem in which the finder cannot sort through large numbers of confusing and ambiguous signals (while also needing to detect stealthy and concealed forces) fast enough to be militarily useful. At baseline, the finder's operational picture already involves an extensive array of possible targets with varying probabilities of location and identification accuracy. The goal of a fog of war machine is to present the finder with the largest possible array of suspected targets while decreasing confidence about the information's quality.

Some deceptive countermeasures are low-tech and do not require AI. These include such measures as attacking and jamming enemy sensors, timing movements, controlling emissions, generating false electronic signals, deploying basic decoys, using operational-level feints, and engaging in tactical-level concealment and dispersion. If militaries take these measures seriously, even lower-tech actors could create significant challenges for an AI information-fusion system.

Other deceptive countermeasures are high-tech, and militaries could leverage AI to enhance them. An AI fog of war machine is planning and battle-management software that could help human operators develop and execute elaborate schemes for deception. Doing so could include integrating the low-tech measures described above alongside additional sophisticated methods, such as operating large numbers of autonomous decoys and conducting cyberattacks designed to either corrupt data or plant misleading information. A fog of war machine could also help make decisions about how to selectively reveal true information in ways that increase collective uncertainty, as well as how to operate real forces in ways that make them look like decoys or that exploit seams in an enemy's sensing networks. Figure 3.2 shows how the use of a fog of war machine might play out during a conflict.

¹⁵ Geist, 2023, p. 128.

FIGURE 3.2
Illustrative Fog of War Machine



NOTE: The arrow connecting the adversary's sensing systems (the box on the left) to the friendly fog of war machine (the middle box) captures the idea that the fog of war machine would consider information about the adversary's sensing networks when planning deception operations. For example, if an adversary's imaging satellite is passing overhead friendly territory, the fog of war machine might tip friendly forces to take cover or queue a counterspace capability to suppress the satellite. In practice, friendly information about the adversary's information-fusion and battle-management systems will be imperfect. Both sides might simultaneously be using fog of war machines to deceive the other, complicating efforts to optimize deception for an enemy's sensing network because that, itself, is subject to deceptive tactics to protect it. ELINT = electronic intelligence.

One potential advantage for the hider is that increasing uncertainty can be easier than resolving uncertainty. The hider's AI's computational task of finding ways to create large amounts of noise and confusion is simpler than the finder's AI's computational task to provide clarity about a complex environment. Analogously, the finder needs to solve a puzzle, and the hider is making that harder by removing some puzzle pieces from the box, adding fake puzzle pieces at the same time, and trying to simultaneously change the picture so that solving the puzzle will result in an outdated solution.¹⁶ Although the finder's sensors and information-fusion system might improve, the adversary can decide in real time on ways to degrade and pollute the view that the finder's information-fusion system sees.

These methods can make it harder for the finder to close kill chains. Again, the attacker needs to accomplish multiple tasks to close the kill chain—find, fix, target, engage, and assess. The hider can use deception to break the find, track, and assess links. For example, the hider can use camouflage and concealment to make it more difficult for an adversary to find its forces; it can use mobility, decoys, and jamming the finder's sensors to make it more difficult to track its forces; and it can use emplaced decoys or spoofed signals to complicate battle damage assessments.¹⁷ Deception is a multilevel game in which multiple players can employ different modes of deception on different scales. Hiders can degrade the finders' operational picture by creating imperfect information, inserting false information, or overloading the system with information.

There are multiple pathways by which an AI fog of war machine could exploit deception for military effect. One approach is to create false information, such as fake forces, to convince the enemy to misallocate its real resources. Allied efforts in Operation Fortitude during World War II exemplify this approach. The Allies created a fake army group to convince the Germans that the primary Allied landing spot for the eventual invasion of France would be along the Pas-de-Calais instead of at Normandy.¹⁸ Decoys, such as inflatable planes and tanks and fake signals traffic, helped accomplish this goal, misleading Hitler into misallocating resources away from the defense of Normandy and supporting the success of the Allied landing in June 1944.

Another approach to deception centers on the use of *real* forces to mislead the adversary. The Soviet Union favored this approach in World War II. Simultaneously with the Allied landing in Normandy, the Red Army embarked on Operation Bagration against the German Army Group Centre. Soviet deception operations convinced Germany that the next Soviet offensive would target Army Group North Ukraine rather than Army Group Centre.¹⁹ This threat was very credible because there were actual Soviet armies posed to attack in Ukraine rather than Belarus.

AI could enhance both approaches to deception and create novel ways to combine them. Moreover, AI stands to make it possible to attempt these kinds of deceptions with previously inconceivable speed and scale. Operation Fortitude required many months of planning and preparation, but, in the future, such efforts might come together in weeks or days.

For leveraging fake forces, AI could facilitate the creation of cheaper, more-realistic decoys. To carry out the deception in Operation Fortitude, the United States created a “ghost army” staffed with actors and other creative types to undertake such tasks as faking plausible signals traffic.²⁰ Today, generative AI could

¹⁶ Geist, 2023, p. 126.

¹⁷ Sherrill Lingel, Matthew Sargent, Timothy R. Gulden, Tim McDonald, and Parousia Rockstroh, *Leveraging Complexity in Great-Power Competition and Warfare: Vol. I, An Initial Exploration of How Complex Adaptive Systems Thinking Can Frame Opportunities and Challenges*, RAND Corporation, RR-A589-1, 2021.

¹⁸ Robert Hesketh, *Fortitude: The D-Day Deception Campaign*, Overlook Press, 2000.

¹⁹ Donald C. Daniel and Katherine L. Herbig, eds., “Deception in Theory and Practice,” in *Strategic Military Deception*, Pergamon Press, 1982.

²⁰ Rick Beyer and Elizabeth Sayles, *The Ghost Army of World War II: How One Top-Secret Unit Deceived the Enemy with Inflatable Tanks, Sound Effects, and Other Audacious Fakery*, Chronicle Books, 2023.

replace thousands of human deception specialists with off-the-shelf software. With advances in robotics and uncrewed systems, the physical aspects of deception operations might also greatly improve. In the years before a conflict, AI could help design forces to enhance deception, including designing new physical and electronic decoys to make them more realistic and capable of operating autonomously, as well as designing new real capabilities to make their signatures more closely resemble the decoys. The practicality and cost-effectiveness of decoys might increase significantly relative to earlier wars.

Similar enhancements in cost-effectiveness might occur for deception using actual forces. When procured in large numbers, uncrewed systems can devote a larger fraction of forces to feints and other forms of deception while still retaining a large capability for the main effort. Inexpensive, autonomous uncrewed platforms might also facilitate combinations of offense and deception that were not practical with past technologies. It would make very little sense to equip an inflatable tank with a real gun. But a drone designed primarily for deception could be equipped with some real weapons (such as a warhead and the ability to switch into attack mode), forcing the adversary to take it seriously even if it knows that the drone is not what it pretends to be. Such combinations of deception and offense could permit novel operational concepts.

An AI fog of war machine is certainly not a silver bullet. Deception involves trade-offs, which is why militaries are not always eager to embrace it. For example, a ship or an aircraft that controls its radar emissions is harder to find but also reduces its own ability to find enemy forces. These kinds of inefficiencies create a tax on a unit's operational effectiveness and could generate compounding effects on military utility. Furthermore, the investments in resources and training create trade-offs for force development, requiring militaries to invest scarce resources in decoys rather than in more-directly usable military capabilities.

Additionally, it will not always be easy to deceive a finder's AI information-fusion system. The finder will attempt to train its AI systems to recognize the hider's deceptive actions. The AI tools used for reconnaissance and counter-reconnaissance are therefore likely to engage in a prolonged cat-and-mouse game, with neither having overwhelming advantages over the other that would be necessary either to lift the fog of war or to completely blind and confuse an adversary. As we detail in the next section, there will also be variation in the circumstances under which hiders can create significant uncertainty for finders.

When Will AI Help Hiders or Finders More?

There are three variables that will shape the conditions under which hiders or finders gain an advantage. First, mass plays a key role. Mass can make deception more effective by creating so much noise that the finder cannot make sense of its environment; however, sufficient mass on the finder's side might neutralize some forms of deception. If the finder floods an area with enough sensors and aggressively destroys anything that might be a real target, then it could end up destroying the relevant targets despite uncertainty about which targets are real.

But there are also constraints to a finder taking a scorched-earth approach that leaves nothing untouched. The finder has a finite inventory of strike platforms and munitions, so using them recklessly could create capacity problems, such as running low on weapons. The hider could also employ defensive interceptors and counter-battery fire to further complicate the finder's ability to execute the kill chain without putting its own forces at risk. Because the hider can shoot back when the finder exposes its forces to attack a decoy, the finder might want some confidence that it is exposing its forces for a good reason. Indeed, the hider-finder dilemma works both ways for a deceiver that has its own long-range strike complex *and* information-fusion system.

Second, the amount of uncertainty the hider can create depends on the kind of information it is trying to conceal. A fog of war machine's greatest utility is creating uncertainty and confusion for the finder. In cases in which the fog of war machine cannot actually hide the real target from the finder's sensors, it can create a kind of shell game so that the finder is uncertain about which targets it senses are the real ones. This can work

for mobile systems, such as road-mobile missile launchers and ships. It will not work for hiding the locations of well-known fixed targets, such as cities or ports. Additionally, this kind of deception works best for creating uncertainty when the requirements for accurate and precise information are highest. It might be quite difficult for the hider to obscure the general position of an infantry brigade. It is more feasible to prevent the finder from collecting targeting intelligence on a specific mobile missile or air defense system.

Although assessing the implications of using AI for nuclear deterrence is beyond this report's scope, we note that preserving a secure second strike is a comparatively easier task for the hider than many other tasks. This is because (1) the finder needs to locate all or nearly all the hider's nuclear forces in a narrow time frame and with a high degree of confidence to enable a successful disarming strike, (2) great powers have a variety of mobile nuclear delivery systems operating in multiple domains, (3) there are strong incentives to invest in deception and defenses to protect those strategic forces, and (4) incremental attrition is of limited utility to the attacker (unlike in conventional contexts).

Third, there is variation across different domains and physical environments. Wider and more-cluttered search areas lend themselves to hiding, such as ground forces operating in urban or forested terrain. Urban terrain has more physical site-blocking structures that hidiers can use to hide from the finder's sensors and more civilian activity that could create false positives, such as uncertainty for the finder about whether it has found a civilian or military vehicle. Smaller and emptier areas lend themselves to finding. Forces operating in narrow geographical areas reduce the search radius and the amount of uncertainty the hider can create. Forces operating in empty geographical areas, such as ground forces operating in the desert or aircraft flying against the uncluttered background of the sky, make finding them easier. Deception is not impossible in these domains because militaries can still employ tools, such as decoys, but it is more challenging.

Practical implications will depend not just on the domain but also on the exact circumstances. For example, the underwater domain mostly favors hiding because of physics, but certain sensors can detect even stealthy craft over short-enough distances. A well-defended maritime chokepoint could therefore prove a formidable obstacle even for sophisticated hidiers. The dynamic relates to relative knowledge-quality problems: A small area, even one in a challenging domain (such as the undersea domain), can be well characterized and therefore favor finders, while a large area in a comparatively simpler domain (such as the air) can be challenging to monitor at scale and therefore favor hidiers at times.

Taken together, there are specific conditions under which hidiers or finders will most likely gain an edge. The hider's job is easiest when the hider can better leverage large numbers of decoys, when it is trying to prevent the finder from having high confidence about information that must be highly precise and accurate, and when it is operating in physical environments that are larger and more cluttered. The finder's job is easiest when the finder can better leverage large numbers of sensors, when its goals require less-precise information and enable greater tolerance for uncertainty, and when it is searching for targets in smaller and emptier physical environments.

Conclusion

Although advances in AI and the proliferation of sensors will enable militaries to conduct more-effective reconnaissance, commentators sometimes overstate the extent of these trends. Militaries will likely invest in deception and counter-reconnaissance to offset improved scouting. Some of these countermeasures will be low-tech, such as attacking sensors and controlling electronic emissions. Others will be high-tech, such as fielding a new generation of autonomous decoys and developing AI tools for deception planning and battle management.

Because of advances in finding, the U.S. military will have to invest far more in deception than it has in the past. This will require a mindset shift from the past three decades in which the U.S. military had

overwhelming advantages over regional adversaries and insurgents and its main challenge was to find targets, not to avoid being found itself. This new emphasis on hiding is necessary, but it will come at a cost. Investing in sophisticated decoys will require allocating scarce resources in the face of competing priorities. Some deceptive tactics, such as limiting movement and controlling emissions, also decrease operational effectiveness and flexibility.

The hypothesis that hiding will become more sophisticated and thereby keep the hider-finder competition contested in the future rests on the assumption that militaries take hiding seriously. Making deception effective can be difficult and costly. If hiders fail to embrace deception and allocate significant resources to it, the prediction might not hold.

Centralized Versus Decentralized Command and Control

How will AI affect the relative advantage between centralized and decentralized approaches to C2? As a technology that can engage in analytical reasoning and decisionmaking, AI has an intuitive appeal for strengthening C2. AI can outperform humans in well-structured strategy games, such as StarCraft, by making much faster decisions and developing novel strategies.¹ Although AI still struggles in more-ambiguous and unstructured strategic settings, such as warfare, advances in AI could help human commanders make better and faster decisions than they could on their own.²

How militaries will adapt their approaches to C2 in response to AI remains an open question. Some military futurists envision AI enabling a more centralized “hive brain” in which commanders can use AI to expand their span of control and manage much larger numbers of forces—and do so at a level of detail that had been impractical in the past.³ Others foresee AI empowering a more decentralized and distributed approach in which forces closer to the tactical edge can use AI to make better decisions and rely less on centralized direction.⁴

We argue that the hybrid C2 approach of *mission command* will remain preferable for the U.S. military, even while AI benefits both centralized and decentralized C2 in absolute terms.⁵ Mission command involves centralized decisionmaking but decentralized execution, dispersing command among multiple echelons and empowering subordinate judgment at all levels about how to execute orders in ways that are best suited to a situation.⁶ Neither more-centralized nor more-decentralized approaches to C2 will supplant mission command because access to information, not just cognitive capacity, is a limiting factor for moving toward different C2 paradigms. There is a key information asymmetry to consider: Theater-level commands have greater awareness of a military campaign’s operational-level environment, while forward commands at lower

¹ Paul Scharre, “AI’s Inhuman Advantage,” *War on the Rocks*, April 10, 2023b.

² On the limitations of existing AI technologies for C2, see Goldfarb and Lindsay, 2021/22; Walsh et al., 2021; and Cameron Hunter and Bleddyn E. Bowen, “We’ll Never Have a Model of an AI Major-General: Artificial Intelligence, Command Decisions, and Kitsch Visions of War,” *Journal of Strategic Studies*, Vol. 47, No. 1, 2024.

³ See, for example, Sydney J. Freedberg, Jr., “Empowered Edge Versus the Centralization Trap: Who Will Wield AI Better, the US or China?” *Breaking Defense*, February 13, 2024.

⁴ Clark, Patt, and Schramm, 2020.

⁵ Our focus is on the U.S. military. On Chinese perspectives, see Nathan Beauchamp-Mustafaga, Ivana Ke, Amanda Kerrigan, and Edmund J. Burke, *Mission Command with Chinese Characteristics? Exploring Chinese Military Thinking About Command and Control in Future Warfare*, RAND Corporation, RR-A2257-1, 2025.

⁶ This definition is an adaptation of the U.S. Joint Chiefs of Staff and U.S. Army doctrine definitions for mission command. See Joint Chiefs of Staff, *Joint Campaigns and Operations*, Joint Publication 3-0, June 18, 2022, p. III-5; Headquarters, Department of the Army, *Mission Command: Command and Control of Army Forces*, Army Doctrine Publication 6-0, July 2019. See also J. P. Clark, Joe Broome, Derrick “Tank” Franck, Jr., and Michael Loftus, *Command in Joint All-Domain Operations: Some Considerations*, U.S. Army War College, July 22, 2020, p. iii.

echelons have greater awareness of their specific tactical situations. Militaries cannot always count on seamless communication during wartime to resolve these information asymmetries within operationally relevant time frames. Mission command maximizes adaptability in the face of contingency and uncertainty by enabling a form of hierarchical comparative advantage, with different echelons best suited to make different decisions using their respective informational pictures and contexts. Advanced AI will not fundamentally change this dynamic, so it does not change mission command's relative advantage. Rather than replacing mission command, AI is more likely to improve C2 practices within this paradigm.⁷

How AI Can Help Both Centralized and Decentralized Command and Control Architectures

Advanced AI has a clear appeal for C2 applications. Modern battlefields can be extremely complex with large numbers of friendly and enemy platforms dispersed across wide geographic areas and in multiple domains. AI could help human commanders better understand their environments and make decisions about how to employ the forces available to them.⁸ AI's performance in strategy games suggests that AI agents could offer benefits in decisionmaking speed, coordination, resource management, and calculated risk-taking.⁹ These benefits could be particularly important for helping humans control large numbers of robotic units that, as we suggest in Chapter 2, could become an increasingly common feature in future battlefields. At the same time, AI has limitations that humans can help offset, and its performance in structured computer simulations does not guarantee that it will perform well in dynamic combat environments with adversaries deliberately trying to break the rules. There are also compelling political, social, and legal reasons for humans to remain in ultimate command of decisions about using force. In this chapter, we therefore assume that the U.S. military will use AI to assist humans in exercising C2 rather than simply delegating command responsibility to AI agents.

Advances in AI could strengthen centralized approaches to C2. By *centralized approaches*, we mean theater-level commands exercising relatively tight and rapid control of forces throughout the theater. One reason to take an echeloned approach to C2, in which commanders delegate the implementation of tasks to lower-level units, is because commanders have limited cognitive bandwidth. They can manage only so much information at once, so they face limitations in how many forces they can directly oversee and control with the speed, adaptability, and sound judgment required for effective decisionmaking. This challenge is rooted in the amount of cognitive bandwidth available to commanders, not the quality of human intelligence or problem-solving skills. Advanced AI could strengthen more-centralized approaches to C2 by mitigating this cognitive capacity challenge. It could do so by analyzing and making sense of large amounts of data for human commanders, rapidly identifying and evaluating potential COAs, and speeding up the process of translating high-level decisions into nested operational taskings at each echelon.¹⁰ The laborious planning process and time required to take these steps in the past meant that human commanders had to rely heavily on decentralized execution because updating orders quickly in dynamic environments was impractical. AI could facilitate more-dynamic adjustments, such as updating Air Tasking Orders, that currently face constraints in human

⁷ Benjamin Jensen and Jake S. Kwon, "The U.S. Army, Artificial Intelligence, and Mission Command," *War on the Rocks*, March 10, 2025.

⁸ As we explained in Chapter 3, AI might also play a major role in supporting deception, but all else being equal, humans will fare better trying to make sense of their environments with advanced AI tools than without them.

⁹ Scharre, 2023b.

¹⁰ Clark, Patt, and Schramm, 2020.

staff's ability to make changes relative to an operation's tempo.¹¹ Such changes would matter the most for particularly complex operations, such as those that require carefully sequenced and synchronized effects across multiple domains. Additionally, AI could change how centralized orders are executed at lower levels. For example, one can imagine that part of the orders process could be training an AI agent on the commander's thinking and the staff's iterative discussions of the planning steps, wargames, and briefs. This "commander's intent" agent could be copied and replicated at subordinate C2 nodes to inform their decisionmaking.

Advances in AI could also strengthen decentralized approaches to C2. By *decentralized approaches*, we mean that tactical command elements close to the forward edge of the battlefield have more flexibility to make decisions about how they operate. AI might be particularly valuable for augmenting smaller command staffs with less planning experience. Forward units could use AI tools to develop COAs informed by fusing data on such factors as the disposition of enemy forces, the terrain, and doctrine.¹² Additionally, AI-enabled autonomy could be particularly important for C2 of uncrewed systems operating in contested electromagnetic environments in which the radio-frequency links required for humans to directly control such systems might be jammed. Similarly, autonomy will be particularly important for managing the operations of drone swarms where airspace deconfliction might be challenging and when there might be more platforms than there are human pilots available to operate them.

Why Mission Command Will Remain Desirable

Mission command is a hybrid of centralized and decentralized C2 approaches. Foundational to mission command is the idea of "centralized control, decentralized execution."¹³ Militaries choose different C2 arrangements in response to trade-offs, and one issue that mission command helps address is having access to the right information at the necessary time and place to make a sound decision. There is a trade-off between the proximity of forces to tactical engagements and their access to different kinds of information (see Figure 4.1 for a visual representation of this trade-off).¹⁴ As distance from the forward edge of the battlefield increases, situational awareness about tactical engagements and what is happening with the forces in contact with the enemy decreases. Put simply, the farther you are from the front lines, the less you know what is happening on the front lines. By contrast, situational awareness about the operational-level environment increases because more attention and different data sources become available. The closer you are to the front lines, the less you know and have time to think about what is happening across the entire theater. Another key information asymmetry that occurs across every echelon is the existence of tacit knowledge. *Tacit knowledge* is the shared contextual knowledge among a staff or local commanders that is rarely explicitly written down. For example, a local commander might know that one subordinate force element is more rested or adept for a particular type of operation.

These information asymmetries mean that neither centralized nor decentralized decisionmakers have access to all the information they need about the best way to operate without relying on the other. Mission command leverages the informational advantages of different echelons. Theater-level commands make operational-level decisions about managing the campaign as a whole. Forward forces make tactical decisions about how to implement centralized directives and achieve local goals in ways that support the broader campaign.

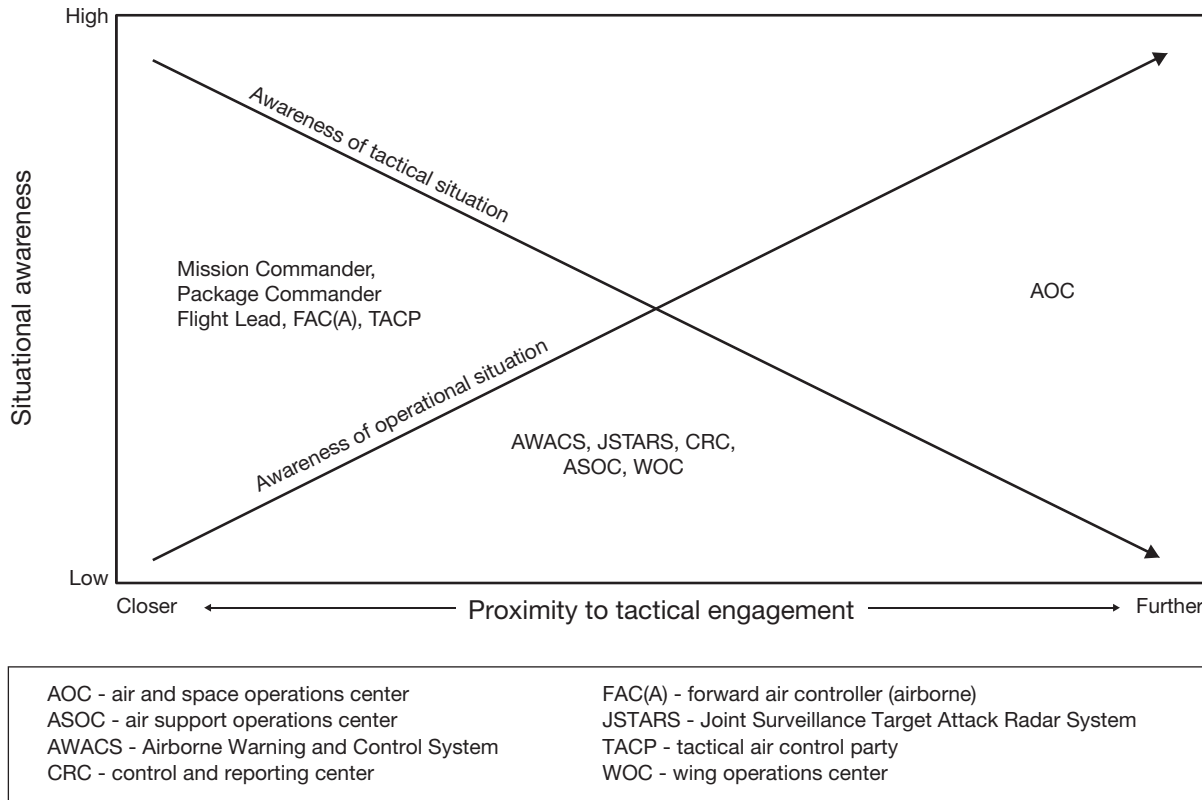
¹¹ Walsh et al., 2021, pp. 4–6.

¹² Jensen and Kwon, 2025.

¹³ Alan Docauer, "Peeling the Onion: Why Centralized Control/Decentralized Execution Works," *Air & Space Power Journal*, Vol. 28, No. 2, March–April 2014.

¹⁴ This draws on Docauer, 2014.

FIGURE 4.1
Situational Awareness Versus Proximity to Tactical Engagement



SOURCE: Reproduced from Docauer, 2014, p. 27.

Existing military C2 constructs evolved in response to these challenges with information asymmetries and cognitive capacity. C2 constructs consist of structure hierarchies (e.g., theater, operational, and tactical echelons) and different functions (e.g., maneuver, fires, logistics, intelligence, protection) that require coordination up and down echelons.¹⁵ Theater commands decompose high-level orders into subordinate tasks that they then assign to a task force, which in turn assigns the tasks to a specific tactical unit.¹⁶ The existing C2 system operates with the expectation that there are significant constraints that prevent theater-level and tactical commands from seamlessly sharing information with each other. Theater commands make decisions about force assignment and positioning orders because they have a better view of the operational-level

¹⁵ Richard L. Daft, *Organization Theory and Design*, 10th ed., South-Western Cengage Learning, 2010.

¹⁶ Take the task of crossing an enemy-defended river as an example to illustrate the value of nested echelon problem-solving. Planning for a river crossing under fire involves many technical details (e.g., the river's depth and speed, the suitability of the crossing site, the quantity and quality of bridging material) and intangible variables (e.g., the subordinate commanders' experience and personalities, cross-unit familiarity and interoperability, and confidence in friendly counter-battery fire capabilities in a new situation). In this case, the division echelon is best positioned to plan a scheme of operations and make critical decisions because of its immediate access to dynamic physical data and its repository of tacit knowledge about subordinate commanders and force elements. Meanwhile, higher echelons at the corps level or theater level can focus on shaping the deep battlespace (by interdicting counter-attacking forces) and setting conditions (prepositioning supplies for reconstitution and maneuver forces for exploitation) so that the division's crossing will be successful. This interplay between centralized and decentralized C2 is what produces mission command—an ensemble of centralized plans, nested decisionmaking nodes, and decentralized execution.

situation and understanding of how best to position and sequence the right units and materials to support the overall campaign plan. Tactical commands make decisions about how to implement high-level orders because they have a better view of the details required to understand their particular tactical situations and because they have more attention and practical expertise than headquarters have available to decide how to solve specific tactical problems that arise.

A key question about how much AI might change this dynamic is how much of the problem is ultimately about information asymmetries versus cognitive capacity. If the problem is mostly about insufficient cognitive capacity and attention to process information and develop a response, then AI could empower much more centralized C2. If there is a more enduring problem with getting the right information to the right place, then mission command will retain one of its core comparative advantages.

We argue that informational challenges will remain an enduring problem even if AI can help address challenges with cognitive capacity. Warfare's inherent unpredictability makes it impractical to predict and orchestrate everything in advance. The battlefield environment is dynamic and can change quickly. Limits on communications bandwidth make it impractical for units to convey every piece of tactically relevant information up the chain of command within a reasonable time frame, so headquarters units might lack timely updates necessary to make sound tactical-level decisions.¹⁷ Theater-level commands might also lack tactical knowledge that can be difficult to observe or communicate clearly, such as an intuitive sense of training proficiency or a unit's level of exhaustion. Establishing reliable communication links can be challenging when facing sophisticated adversaries because the enemy can attack communication networks, such as by jamming datalinks. It is difficult to ensure the high-bandwidth, low-latency communications necessary for a centralized battle manager to maintain a comprehensive, "panopticon-like" view of tactical engagements. For human operators, this limitation typically pushes individuals to rely on a combination of preplanned battle drills and real-time tactical adaptations.

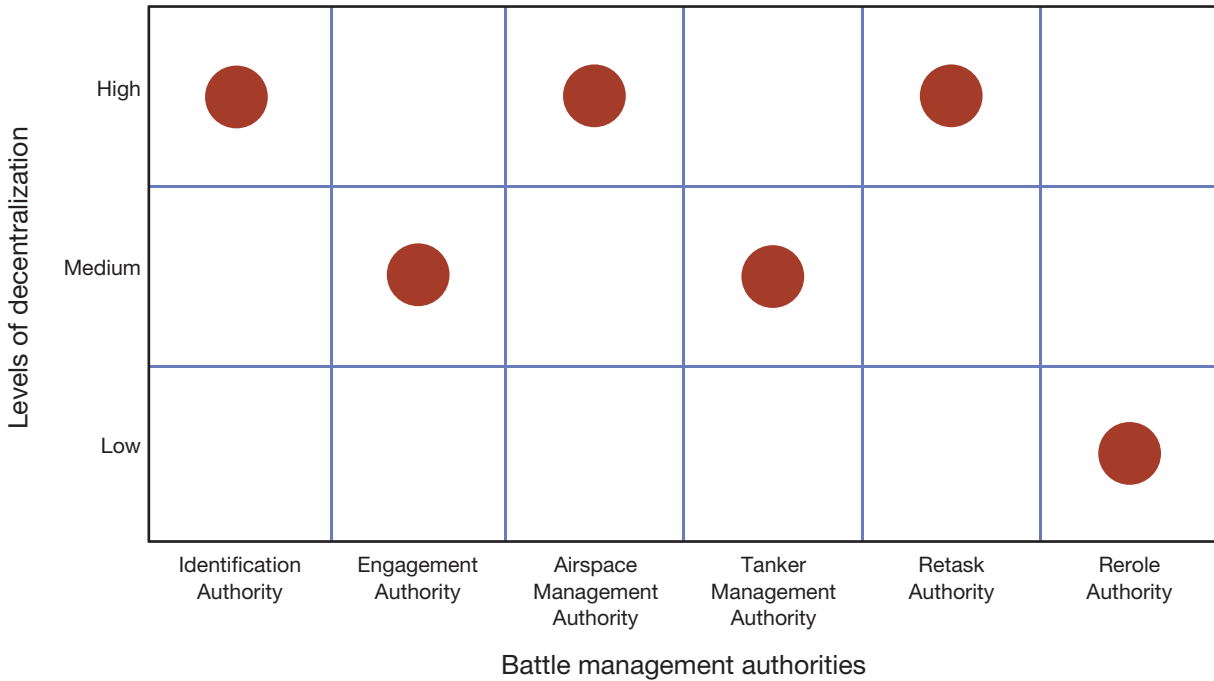
Central to this argument is that operational demands should shape C2 structures, not simply whether a human or an autonomous robot has the cognitive ability to perform a particular task. Commanders prefer to delegate many decisions to those with the best immediate information who can also act quickly in degraded communication environments. Figure 4.2 visualizes this; commanders delegate tactical tasks, such as identifying targets and deconflicting local airspace, while retaining control over bigger-picture decisions, such as what roles certain aircraft or units are supposed to play to support the overall operation. Autonomous agents would face similar issues and would likely settle into relatively similar C2 patterns as crewed systems simply because some of the underlying problems are similar. This is why it is important to think about how C2 structures for autonomous, proliferated agents would respond to operational demands, not just whether an AI-piloted system has the ability to carry out a task on its own.

Even if mission command remains desirable, that does not mean that AI will produce no meaningful changes to C2 practices. We expect that advances in AI will change the way that militaries *implement* mission command. The most significant attainable change to mission command might be that AI-enhanced C2 will enable more *dynamic control* by helping manage such tasks as battlespace deconfliction, logistics coordination, and kinetic and nonkinetic effects synchronization. To manage complexity, mission command has typically relied on rigid mechanisms of predetermined control, such as preplanned operational schemes, fixed unit-coordination boundaries, or dedicated resource allocations. In the future, AI might help make real-time adjustments and reduce frictions that would have previously prevented *in situ* adaptations of complicated coordination schemes of control. These adjustments could support more-dynamic and more-effective tactical responses in such areas as electronic warfare, airspace and fires deconfliction, and maneuver tactics.

¹⁷ Walsh et al., 2021, p. 19.

FIGURE 4.2

Command and Control Decentralization for Different Types of Decisions



SOURCE: Reproduced from Docauer, 2014, p. 40.

Will AI-Enabled Swarming Change Command and Control Paradigms?

One topic that deserves special attention is how our arguments about C2 apply to AI-enabled robotic swarms.¹⁸ *Swarms* are collections of autonomous agents, and *swarming* involves those agents coordinating through emergent behavior to achieve goals in ways that any individual agent might not decide or even fully understand.¹⁹ Tactical tasks for which militaries might be able to use decentralized swarming C2 include distributed sensing and communications, strikes with swarms of one-way attack munitions, and adaptive electronic warfare.²⁰ This swarming concept could involve an alternative C2 paradigm that involves a one-to-many relationship that replaces the nested hierarchies of one-to-few relationships that exist in mission command. Swarming could make uncrewed systems more adaptable and less dependent on centralized C2 for tactical problem-solving, potentially enabling novel tactics in unpredictable environments.

However, swarming is still compatible with mission command approaches to C2. A human decisionmaker can establish a central goal for the swarm, and the swarm can then determine a decentralized approach to

¹⁸ On swarms, see Paul Scharre, “Unleash the Swarm: The Future of Warfare,” *War on the Rocks*, March 4, 2015; Paul Scharre, *Robotics on the Battlefield Part II: The Coming Swarm*, Center for a New American Security, October 2014; John Arquilla and David Ronfeldt, *Swarming and the Future of Conflict*, RAND Corporation, DB-311-OSD, 2000; and Sean J. A. Edwards, *Swarming on the Battlefield: Past, Present, and Future*, RAND Corporation, MR-1100-OSD, 2000.

¹⁹ Robert E. Price, “Defining Swarm: A Critical Step Toward Harnessing the Power of Autonomous Systems,” *Military Review*, May 2025; Ross Arnold, Kevin Carey, Benjamin Abruzzo, and Christopher Korpela, “What Is A Robot Swarm: A Definition for Swarming Robotics,” *2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, Institute of Electronics and Electrical Engineers, October 2019.

²⁰ Scharre, 2015.

pursuing that goal. The tactical-level C2 within the swarm might look quite different, but the overall construct for how the U.S. military would operate would still prioritize mission command and find ways to integrate swarms into that construct.

Additionally, the effectiveness of swarming still depends partly on the reliability of communication and the availability of information. The performance of swarms depends on the ability of agents within the swarm to pass information to each other and coordinate their efforts. Even small amounts of timely, relevant information shared between networked agents in a swarm can lead to significant performance gains. Past RAND research found that increasing communication between uncrewed aerial vehicles in a swarm has a dramatic impact on the swarm's ability to find and kill targets, and increases in intraswarm communication have proven more important for combat outcomes than increases in military intelligence.²¹ For example, agents in the swarm can share information with each other on a target's location to offset challenges with target location errors, and they can better respond to attrition by repositioning agents to maintain coverage.²² Even in the case of swarming, therefore, access to information rather than intelligence alone is still a critical enabler of effective decisionmaking.

Conclusion

Advanced AI will reshape C2, but its impact will be more evolutionary than revolutionary. AI's likely benefits include automating a great deal of staff work that is currently laborious and time-consuming, potentially enabling humans to make better and faster decisions. Improving the backbones of C2 systems, including communications networks and military intelligence analysis to make sense of the environment, will help with that process. AI might also let people command autonomous platforms in ways that resemble how people currently command other people, making it practical to employ uncrewed systems both at unprecedented scale and in novel ways.

However, AI is unlikely to change the informational constraints that make mission command preferable to significantly more-centralized or more-decentralized approaches to C2. Human and machine intelligence both face the same challenge in access to the right information at the right place and time. Centralized and decentralized C2 arrangements each offer value because different actors have access to different information. Theater-level commands have a better picture of the overall operational environment, and tactical commands have a better picture of what the forward edge of the battlefield looks like. Mission command will continue to work well because it combines the best of both approaches, allowing each echelon to make decisions best suited to the informational picture it has available. Removing the limitations of human cognitive capacity on military operations does not fundamentally change this dynamic.

²¹ When platforms share small amounts of information and adapt tactically based on common behaviors, the swarm achieves better outcomes with fewer systems, simpler sensors, and limited onboard decisionmaking (see David R. Frelinger, Joel Kvitky, and William Stanley, *Proliferated Autonomous Weapons: An Example of Cooperative Behavior*, RAND Corporation, DB-239-AF, 1998, pp. 41–43).

²² Frelinger, Kvitky, and Stanley, 1998.

Cyber Offense Versus Cyber Defense

How will AI affect the relative advantage between cyber offense and cyber defense? AI will improve both in absolute terms, but we argue that AI will likely create greater marginal benefits for cyber defenders over the long term. The key reason is that AI could help address three factors that currently limit cyber defense capabilities: speed, scale, and effectiveness. Defenders face resource constraints that limit their ability to discover vulnerabilities and defend networks relative to the size of the attack surface, and this gives attackers an advantage.¹ AI's potential to (1) enable defenders to produce less error-prone code in the first place, (2) scan defenders' software for vulnerabilities faster and more effectively than they could previously, and (3) develop and implement security patches more rapidly and cheaply could disrupt the economics of cybersecurity in a way that generally benefits the defender.² Defenders could use AI to speed and scale up these tasks in ways that erode the structural advantages that attackers enjoy.³

We hypothesize that cyber defense will benefit more than cyber offense “in the limit”—that is, in the long run and as the technology matures and diffuses. This prediction is conditional on defenders taking advantage of the opportunities that AI provides by investing adequately in AI-enabled defenses. In the short term, if cyberattackers leverage AI tools effectively but defenders do not, the offense's existing advantages could grow even larger.⁴ But, in a long-term equilibrium in which both sides leverage the technology with equal skill, we predict that advances in AI will benefit the defender relatively more than the attacker because AI can help offset some of the weaknesses of existing cyber defense methods and shift the underlying cost-effectiveness of the competition. This does not necessarily mean that the defender will gain an advantage over the attacker

¹ For an overview of debates about the offense-defense balance in cyberspace, see Rebecca Slayton, “What Is the Cyber Offense-Defense Balance? Conceptions, Causes, and Assessment,” *International Security*, Vol. 41, No. 3, Winter 2016/17.

² Chad Heitzenrater, *The Winning Economics of Cybersecurity in an Age of Advanced Artificial Intelligence*, RAND Corporation, PE-A3691-11, August 2025.

³ Other research has assessed the implications of AI for the offense-defense balance, but the analysis tends to focus on the short-term implications of AI, whereas our hypothesis is about the long-term equilibrium of AI diffusion. Some analysts argue that neither the defender nor the attacker will gain an inherent advantage (see Jenny Jun, “How Will AI Change Cyber Operations?” *War on the Rocks*, April 30, 2024; and Andrew J. Lohn, *The Impact of AI on the Cyber Offense-Defense Balance and the Character of Cyber Conflict*, arXiv, arXiv:2504.13371, April 17, 2025). Others argue that AI has, so far, benefited cyber defense, but that benefit could shift to the offense because of three factors: the relative compute costs for focused attacks being lower than the costs for defending an entire attack surface, the ability to increase the tempo of cyberattacks (if AI is able to automate the entire cyber kill chain), and the ability for attackers to tolerate system failure that defenders cannot (see Caleb Withers, *Tipping the Scales: Emerging AI Capabilities and the Cyber Offense-Defense Balance*, Center for a New American Security, September 23, 2025). Consistent with our hypothesis, a third camp argues that the advantage balance depends on factors' different timing. For example, after the release of new software, an attacker might have a short-term edge because AI allows the attacker to more quickly detect vulnerabilities, but the defender might benefit because, given enough time, they can use AI to more easily find and patch all vulnerabilities (see Ben Garfinkel and Allan Dafoe, “Artificial Intelligence, Foresight, and the Offense-Defense Balance,” *War on the Rocks*, December 19, 2019b). It is worth noting that a core recommendation from each of these arguments follows our own: that a concerted investment in cybersecurity is warranted.

⁴ For an overview of early empirical examples of cyberattackers using AI to support their operations, see Google Threat Intelligence Group, “Adversarial Misuse of Generative AI,” Google Cloud, January 29, 2025.

in absolute terms, although that is possible under certain conditions. A shift in relative cost-effectiveness also does not mean that cyberattacks will become impractical or unaffordable. Cyberattacks will remain an important challenge for AI in the future, but there are underappreciated opportunities to leverage AI to shift the competition onto more-favorable terrain for the defender.

To make this argument, we break down the cyber offense versus cyber defense building block competition into three component parts—vulnerability detection, maneuver and effects detection, and patching versus exploitation—and analyze how AI might reshape the dynamics of those competitions.⁵

How AI Can Help Both Attackers and Defenders Find Cyber Vulnerabilities

Gaining access to computer networks first requires an attacker to identify vulnerabilities in the defender's systems. Consider a basic model focused on software vulnerabilities.⁶ Each system has N lines of code that contain V vulnerabilities, which are errors in the programming that an attacker can exploit. The number of vulnerabilities in a system increases as the system becomes more complex because having more lines of code will tend to create more opportunities for errors, and some subset of those errors will result in vulnerabilities. Both the attacker and the defender are searching for vulnerabilities, which the attacker wants to exploit before the defender can find and patch those vulnerabilities. For the purposes of the model, assume that both sides are independently searching for vulnerabilities so that each side is essentially making a random draw from the set of V vulnerabilities that exist at any given time. The chance that the attacker or defender finds a given vulnerability is therefore $1/V$ for each side. The chance that they both find the same vulnerability is $1/V^2$. When V is large, the probability that the defender finds the same vulnerability as the attacker is low. Closing the gap between what the attacker and defender will both find requires the defender to allocate vastly disproportionate resources as V grows larger. This dynamic contributes to the attacker having a structural advantage over the defender in vulnerability detection.⁷ Figure 5.1 visualizes this advantage. The shaded gray region below the curved line shows the difference between (1) the probability that the attacker or the defender finds a given vulnerability and (2) the probability that both find the same vulnerability.

AI could help the defender produce less error-prone code that has fewer vulnerabilities, thereby reducing the size of the attacker's asymmetric advantage. If the number of vulnerabilities (V) decreases, then the chance that the defender finds the same vulnerability as the attacker increases ($1/V^2$). There are two opportunities to find and address vulnerabilities: making pre-deployment fixes to software before it is fielded and making post-deployment patches to correct lingering errors.

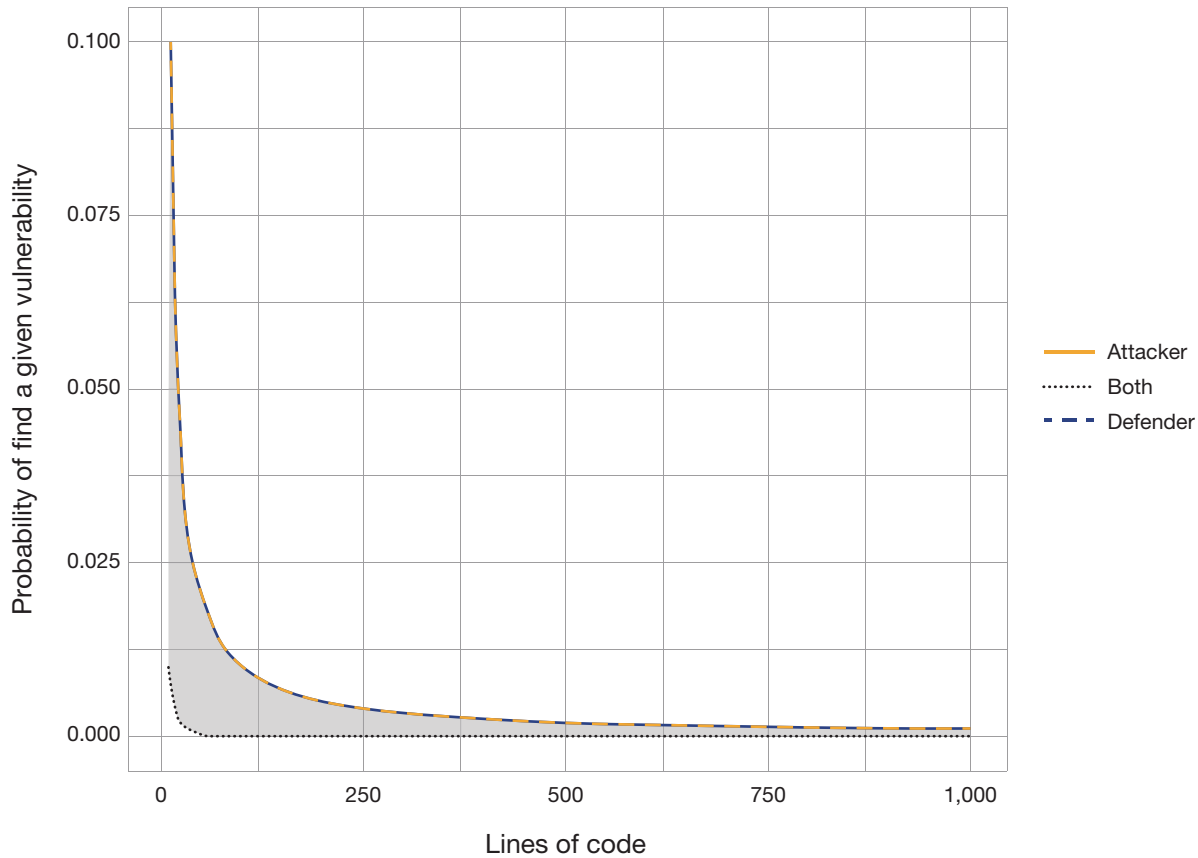
In the pre-deployment stage, AI could help the defender find and fix vulnerabilities before the attacker has a chance to start searching for them in the first place. This is a potential source of asymmetric advantage for the defender. A defender effectively using AI to produce more-efficient code that is less prone to errors during the development stage could significantly reduce the size of the attack surface. The general rule of

⁵ For a similar approach, see Andrew Lohn and Krystal Jackson, "Will AI Make Cyber Swords or Shields?" *Center for Security and Emerging Technology*, August 2022.

⁶ This model of software vulnerability is based on Ross Anderson, "Why Information Security Is Hard – An Economic Perspective," in *Seventeenth Annual Computer Security Applications Conference*, IEEE, December 13, 2001. For an alternative model, see Ben Garfinkel and Allan Dafoe, "How Does the Offense-Defense Balance Scale?" *Journal of Strategic Studies*, Vol. 42, No. 6, 2019a.

⁷ Anderson, 2001.

FIGURE 5.1

Visualizing an Attacker's Structural Advantage in Vulnerability Detection

thumb is that there will be three exploitable vulnerabilities per 10,000 lines of code (or 10 KLOC).⁸ If the quality of AI tools for software development significantly improves in the future, V could start to decrease, and the attacker's current structural advantage would erode. Lowering the average rate of vulnerabilities to three per 100 KLOC or 1,000 KLOC, for example, would make the defender's job significantly easier. The attacker might always have some nonzero number of vulnerabilities that it could exploit, even if only because of user errors or problems with the network's configuration, but the size of the potential attack surface could fall dramatically relative to what attackers enjoy today.

Again, an important caveat is that our hypothesis is about shifts in relative advantages as AI matures and diffuses. In the short term, the dynamics of the relative advantage between the cyber defense and cyber offense will be more complicated.⁹ AI is already playing an increasingly central role in software development. In 2023, 92 percent of software developers in the United States reported using AI coding tools.¹⁰ In 2024, Google claimed that it had used AI to generate more than 25 percent of its new code.¹¹ But early research

⁸ Mark Sherman, "Using ChatGPT to Analyze Your Code? Not So Fast," *Software Engineering Institute Blog*, Carnegie Mellon University, February 12, 2024. KLOC is shorthand for "thousand lines of code."

⁹ Richard Danzig, *Artificial Intelligence, Cybersecurity, and National Security: The Fierce Urgency of Now*, RAND Corporation, PE-A4079-1, July 2025.

¹⁰ Benj Edwards, "Google CEO Says over 25% of New Google Code Is Generated by AI," *Ars Technica*, October 30, 2024.

¹¹ Edwards, 2024.

suggests that AI coding assistants and AI-generated code could increase rather than decrease vulnerabilities in the short term, especially when implementation is poor.¹² Beyond simply generating code with mistakes, the AI coding tools are creating new categories of problems: For example, AI sometimes hallucinates the names of software packages, so malicious actors have created fake packages with these names in anticipation of human programmers installing the AI-recommended packages without verifying their authenticity.¹³ But if the technology continues to advance and humans learn to use it more safely and reliably, these problems might decrease after what could be a painful transition period. In the long run, it is even possible that AI agents will eventually write code and discover vulnerabilities at an expert level that outperforms most humans.¹⁴ This could dramatically reduce pre-deployment vulnerabilities relative to their frequency today.

In the post-deployment stage, both the attacker and the defender can use AI to scan for vulnerabilities. The attacker still has an advantage by only needing to find one vulnerability to exploit, which is why it is so important for the defender to address vulnerabilities in the pre-deployment stage. Both attackers and defenders already use automated tools for vulnerability detection, but those automated tools generally still lag behind what expert humans can accomplish, for now.¹⁵ The attacker's ability to scale up network reconnaissance could mean that a larger number of networks will be under threat compared with attackers' existing capacity. While attacking any one security-hardened network might become more difficult as well-resourced defenders gain better tools, attackers might be able to use AI to attack much larger numbers of non-military targets, which might receive less attention and fewer resources regarding cybersecurity.¹⁶ Militaries reaching even further into their adversaries' societies and domestic critical infrastructure could still have military significance because of each side's ability to impose larger economic and societal costs on the other during wartime, even if most military networks are adequately hardened against attack.

How AI Can Help Detect and Hide an Attacker's Maneuvers and Effects

Beyond finding vulnerabilities to enter the defender's networks, the attacker usually needs to maneuver within those networks once it has access. Although not all attacks require extensive persistence or movement in a system, most involve a series of steps to place the attacker into position to generate a desired effect.¹⁷ If

¹² Jessica Ji, Jenny Jun, Maggie Wu, and Rebecca Gelles, *Cybersecurity Risks of AI-Generated Code*, Center for Security and Emerging Technology, November 2024; Neil Perry, Megha Srivastava, Deepak Kumar, and Dan Boneh, "Do Users Write More Insecure Code with AI Assistants?" arXiv, arXiv:2211.03622v3, December 18, 2023.

¹³ Sarah Gooding, "The Rise of Slopsquatting: How AI Hallucinations Are Fueling a New Class of Supply Chain Attacks," *Socket*, April 8, 2025.

¹⁴ Lohn, 2025, pp. 11–12.

¹⁵ This is a fast-moving area of research, and vulnerability discovery encompasses a broad set of activities. Research points to such factors as the coding language used, the root causes of the vulnerabilities, and the novelty of the vulnerabilities as driving the AI tools' abilities in discovering vulnerabilities. Improvements have come from using AI agents to orchestrate existing code analysis tools rather than relying solely on large-language model code analysis, and varying degrees of human and AI collaboration have provided the best results. See Lohn and Jackson, 2022, pp. 8–9; Maia Hamin, Jennifer Lin, and Trey Herr, "AI in Cyber and Software Security: What's Driving Opportunities and Risks?" Atlantic Council, August 19, 2024; and Chris Rohlf, "AI and the Software Vulnerability Lifecycle," Center for Security and Emerging Technology, August 4, 2025. Notable examples of these capabilities include the AI Cyber Challenge, or AIxCC, from the Defense Advanced Research Projects Agency (DARPA), and Google's CodeMender (DARPA, "Artificial Intelligence Cyber Challenge," homepage, undated; Raluca Ada Popa and Four Flynn, "Introducing CodeMender: An AI Agent for Code Security," October 6, 2025).

¹⁶ Jun, 2024.

¹⁷ For example, even an attacker using stolen credentials to access a system might have to elevate privileges (that is, gain the ability to act as a user with greater access) or navigate restricted portions of the system, and these steps provide opportuni-

the defender can detect the attacker's maneuvers or the effects that its operations are creating, the defender can become aware of the breach and take countermeasures. The attacker, therefore, wants to leverage stealthy cyber tactics to mask its operations, and the defender wants to detect the attacker's maneuvers and effects.

AI could help defenders better monitor their networks by improving defenders' baseline understanding of their own domains. Following the logic above that AI's largest benefits to the defender will be in the pre-attack phases, AI could strengthen the defender's ability to model and better understand its networks in ways that will bolster its ability to respond to attacks later. Defenders often do not have a sufficiently accurate or complete view of their own networks because artifacts, such as network diagrams, can quickly become dated as networks evolve, and such normal behaviors as installing new hardware, patching software, and conducting general maintenance mean that changes to the network accumulate over time. These issues might lessen the defender's understanding of its own network and create challenges in mounting an effective defense once an attacker targets the network. A focused attacker that conducts effective reconnaissance might even have a better view of the target's posture than the defender itself. As with the resilience measures discussed previously, AI tools might enable cyber defenders to keep track of their own postures much more quickly, cheaply, and effectively, thereby increasing the defenders' preparedness to engage attackers once they detect any attackers on their networks.

Another argument often made in favor of the defense is that an improved baseline understanding of the defender's networks could increase the defender's ability to detect attackers if it has a stronger sense of what normal network activity looks like. This knowledge enables the defender to search for signs that the attacker has gained access. The defender can do this by monitoring its networks and searching for anomalies, such as a certain part of the network activating at irregular times. The defender could leverage AI's ability to monitor the defender's networks over long periods and process increasingly large volumes of data to create an AI-enhanced picture of baseline network activity. Doing this can help the defender identify deviations from the baseline by combining anomaly detection with known patterns of an adversary's operations. In theory, AI could give the defender an advantage by monitoring and analyzing networks at a speed and scale that is impractical or unaffordable for human personnel. In practice, however, these approaches have limitations. Anomaly detection has long been a research topic in cybersecurity, and researchers have found mixed results.¹⁸ AI-enabled anomaly detection in complex environments, especially those that might arise during crises or conflicts, has proven especially challenging; by their nature, such environments do not conform to past patterns of behavior, so everything might start to seem anomalous.¹⁹ False positives in such situations might be particularly problematic and divert the defender's resources toward nonthreats. Additionally, attackers will use the same technology to attempt to mask their maneuvers on a defender's network, learning from past successful attacks while also adjusting tactics, making it more difficult for AI cybersecurity tools to anticipate new approaches. Finally, such approaches are necessarily reactive, and in a combat environment (in which the attacker's objectives might call for swift action to deny access to a system, as opposed to

ties for detection. However, in any situation in which attribution or long-term access is not important to the attacker (such as during hostilities), this period in which detection is possible might become vanishingly small. This issue is a major challenge for modern cyber defense, and it supports a strong argument for a focus on proactive resilience.

¹⁸ Joshua Steier, Erik Van Hegewald, Anthony Jacques, Gavin S. Hartnett, and Lance Menthe, *Understanding the Limits of Artificial Intelligence for Warfighters: Vol. 2, Distributional Shift in Cybersecurity Datasets*, RAND Corporation, RR-A1722-2, 2024.

¹⁹ On the other hand, over the very long term, defenders might also gain experience from navigating their environments during irregular periods, such as crises or conflicts, and use that experience to improve their understanding of their networks. That is, defenders might eventually have data about what their networks usually look like even during these anomalous periods.

cyberattacks for espionage focused on data confidentiality), there are good reasons to believe that a detect-and-react approach might be insufficient to prevent many military cyber harms.

As a result, the relative advantage that AI offers the attacker versus the defender in this sub-competition between detecting and hiding cyber maneuvers is less clear than with vulnerability detection. It will likely remain a cat-and-mouse game. There is some reason to believe that AI might benefit the defender in the longer term—but only if the defender leverages its home-field advantage effectively and uses AI to better design and understand its own networks during the pre-engagement phase. The more efficiently and securely the defender can use AI tools to reduce its attack surface and understand its operational environment, the harder and more complicated it will be for the attacker to stealthily operate on the defender's network for long periods.

How AI Can Help Patch and Exploit Cyber Vulnerabilities

The attacker still needs to exploit vulnerabilities after it finds them, and the defender still needs to design and deploy patches to fix vulnerabilities. The delay between detecting a vulnerability and patching it gives the attacker a potential window of opportunity to achieve its desired objectives. How quickly the attacker can do so depends partly on whether it needs to develop new code or whether existing off-the-shelf tools work for exploiting a specific vulnerability.

AI could benefit cyber defenders by speeding up their reaction times once they discover vulnerabilities. On average, defenders are already slightly faster at writing patches than attackers are at developing exploits, but defenders face limitations in the time it takes to deploy those patches.²⁰ Implementing a fix is by far the most expensive and time-consuming challenge for the defender. The defender has to not only develop the fix but also ensure that the update does not inadvertently cause other problems; for example, the cybersecurity company CrowdStrike released an update in 2024 that interfered with other critical functions, resulting in millions of computer outages globally.²¹ AI has the potential to execute this cycle at a much faster pace.²² In addition to developing the patches themselves, AI could speed up implementation by testing how patches interact with other critical software, generating the necessary documentation for the change (e.g., requests, approvals, and records), and helping maintain configuration plans. These are all tasks that make continuous defense so difficult today, and AI could reduce these burdens that slow the tempo and reduce the scale at which humans operate. Although the attacker might still retain a structural advantage in the probability of finding any one vulnerability before the defender, speeding up the development and deployment of patches could further offset that edge and enable the defender to mitigate the effects of vulnerabilities that do occur.

Conclusion

Although commentators often focus on how AI will create new cybersecurity risks and advantages for the offense, the defender might stand to gain the most from breakthroughs in AI. AI could help defenders find vulnerabilities more quickly, at a larger scale, and potentially more accurately than humans. It could enable faster responses to and analysis and forensics of attacker movements, as well as the operationalization of lessons learned across more systems, more quickly. AI could also automate the generation of patches; the

²⁰ Lohn and Jackson, 2022, p. 11.

²¹ Lohn, 2025, p. 31.

²² DARPA, "AI Cyber Challenge Marks Pivotal Inflection Point for Cyber Defense," August 8, 2025.

testing, verification, and validation of the system in which the fix is required; and the necessary artifacts for reaccreditation and use. Taken together, these uses of AI could help address challenges that currently hold the cyber defense back today.

Critically, this argument does not rest on a particular definition of AGI and is likely to be true for many different kinds of AI in the future. Whatever the specific capabilities are of future AI tools, the ability to affect the speed, scale, and effectiveness of actions in the cyber domain presents a greater opportunity to the defense because of the asymmetric benefit that attackers enjoy in choosing the time, target, and intensity of attacks. Although emergent AI capabilities have potential offensive uses, they fail to address the core challenges that attackers face: executing with precision and novelty, tied to a desired effect. Applying AI to the cyber resilience principles of prevention, mitigation, recovery, and adaptation constrains attacker options and raises the stakes for an attack, providing defenders with a pathway to exercising control of the playing field, which, to date, has been limited by precisely what AI is poised to overcome: human action being too slow, the scope of vulnerabilities being too big or complex, and the technologies for securing systems being too imprecise.²³ It is for these reasons that, in the long run, the defense will likely benefit relatively more than the offense. However, this will only occur if militaries and companies fully embrace the defensive potential of AI technologies and commit to adopting them.

Although AI might erode the size of the attacker's structural advantage, this does not mean that cyberattacks will be impractical or that all networks will be fundamentally secure. AI will also create new kinds of vulnerabilities, and militaries need to prepare for scenarios in which cyberattackers try to seize control of the AI systems that the defenders build to protect their own networks, even as defenders work to minimize the introduction of new sources of vulnerability. For example, if the defender creates an AI super user that has authorities to observe and change every element of the network, then an attacker's ability to control that system could be catastrophic. That is, the AI itself becomes part of the attack surface.²⁴ For many of the applications that we discuss here, defenders could use AI in ways that are not accessible to attackers (e.g., during engineering), while other applications (e.g., AI monitors operating on the network) might result in uses that expose the AI to a direct cyberattack. Ensuring that AI agents and systems are safe and secure will be paramount to a successful deployment.²⁵ AI-enabled cybersecurity and resilience applications will require rigorous oversight in their design and use, especially in their early stages, to realize the upsides of this technology while limiting risk.

²³ For a comprehensive overview of this argument and recommended policy actions, see Heitzenrater, 2025.

²⁴ The same might also be true for defenders targeting the AI systems that attackers use for offensive operations.

²⁵ This will require accounting for not only traditional forms of vulnerability but also entirely new attack classes outside current cybersecurity practices, such as data poisoning, jailbreaking, or even the manipulation of physical objects. See Ram Shankar Siva Kumar and Hyrum Anderson, *Not with a Bug, but with a Sticker: Attacks on Machine Learning Systems and What to Do About Them*, Wiley, 2023.

Conclusion

In this report, we looked at four building block competitions to explore how advanced AI could influence military operations. Our goals were to set the terms of a more systematic debate about how AI could reshape how militaries operate and to generate initial hypotheses about areas of relative advantage. Our four hypotheses are that AI will (1) increase the cost-effectiveness of larger numbers of good-enough uncrewed systems compared with smaller numbers of exquisite systems, (2) necessitate greater investments in and new approaches to deception to offset improvements in finding, (3) not change the advantages of mission command over more-centralized or more-decentralized C2, and (4) make the cyber defense more cost-effective by addressing some of the challenges that currently limit the cyber defense relative to the cyber offense. **Taken together, our analysis suggests that the U.S. military should look for ways to use AI in capabilities and operational concepts that leverage mass, deception, mission command, and cyber resilience.** Because of the deep uncertainty concerning how AI and other technologies will continue to develop in the coming decades, these are tentative hypotheses.

Summary of Predictions

First, AI could give quantity an advantage over quality. The core logic is that AI will enable militaries to field larger numbers of good-enough uncrewed systems at a cost advantage over exquisite systems. AI's ability to make uncrewed systems more capable is significant on its own, but it is the combination of precise mass with affordable mass that will change the underlying cost-effectiveness of quantity versus quality and have the most-significant implications. Exquisite platforms and weapons will still have a role in future conflicts, but militaries that shift toward greater numbers of attritable uncrewed systems will likely gain a significant advantage.

Second, AI could help the hider orchestrate deception efforts that offset AI-enabled advances in finding. The core logic is that finders face limits in the extent to which AI sensor fusion can create a transparent battlefield; hidere, meanwhile, have opportunities to leverage AI to aggravate these challenges for the finder and keep the scouting competition contested. Reasoning under uncertainty is structurally challenging for both humans and machines, and advances in AI do not necessarily resolve this problem if hidere can increase the signal-to-noise ratio in the information that the finder's AI is trying to process. Whether the hider or finder ultimately has the edge in this competition will depend heavily on variables that include each side's ability to leverage (1) mass in particular times and places, (2) the kind of information that both sides are trying to uncover or conceal, and (3) the physical domain and operating environment. The overall result will likely be a contested measure-countermeasure race between hidere and findere.

Third, AI is unlikely to displace mission command with highly centralized or decentralized approaches to C2. The core logic is that a key factor in discouraging more-centralized or more-decentralized C2 models is access to the right information, not just access to more brainpower. Forward and rear command elements often have access to different types of operational and tactical data that they need to make different kinds of decisions. AI will support more-effective C2, but it is more likely to reinforce mission command as a C2 paradigm than to replace it.

Fourth, AI could benefit the cyber defense more than the cyber offense. The core logic is that AI can help address some of the limitations in scale, speed, and effectiveness that currently hinder defenders and give attackers an edge. The task of defending very large attack surfaces against highly skilled attackers that can choose how and where to attack places a large burden on the defenders, which requires investing disproportionate resources in cyber defense to offset the attackers' advantage. The most important change that AI could generate for cybersecurity would be enabling defenders to produce much more secure code in the first place, which would reduce the size of the exploitable attack surface and reduce the extent of attackers' structural advantage. Cyberattackers will also benefit from using AI to support their operations, such as finding vulnerabilities and developing exploits, so attackers will retain the ability to penetrate networks at certain times and places, especially those networks that do not adequately invest in AI-enabled defenses. But over the long term, AI could reduce the magnitude of the attacker's advantages relative to the balance today if militaries seriously commit to leveraging AI for cybersecurity.

Implications for the U.S. Military

If mass and deception will be central elements in future wars, the United States will need to make significant changes to its existing force structure and operational concepts. Since the end of World War II, the U.S. military has prioritized quality over quantity and treated deception largely as an afterthought.¹ How might the United States redesign its force structure using the principles of mass and deception?

U.S. force planning should shift toward a greater emphasis on mass while retaining a high-low mix. Embracing mass will require U.S. force planners to shift from focusing on small numbers of exquisite systems to developing a wider range of attritable and expendable uncrewed systems acquired in larger numbers. Instead of focusing on one-to-one replacements, the United States should examine how combinations of different systems can provide greater cost-effectiveness than one system alone. This will require a mindset shift in terms of how the U.S. military thinks about what attributes to prioritize in the procurement process. For lethality, a weapon's individual "probability of kill" might become less important than a salvo's probability of kill. For example, one exquisite cruise missile might be more capable than any single one-way attack drone, but a salvo of ten AI-enabled drones might have a comparable battlefield effect at a fraction of the price.² For survivability, the United States will need to accept decreased performance and increased attrition of uncrewed systems if it wants to lower costs and free up resources to build a larger force. This will require defense planners to resist the temptation to insist on performance and capabilities that mimic those of exquisite crewed platforms because doing so could negate the cost savings. However, investing in uncrewed systems that have lower costs relative to exquisite systems does not mean that all new capabilities should be cheap in absolute terms. Additionally, the United States should not abandon exquisite capabilities entirely, especially in areas in which it still has large advantages, such as undersea warfare. This dynamic will create tensions between continued investments in exquisite systems and new investments in mass. Further research is necessary to determine what mix of attributes and performance requirements an AI-enabled force should have for different mission areas.

U.S. force planning should incorporate new requirements for deception. The United States will need to change its approaches to force planning and procurement to better leverage deception. The U.S. approach has often been to design its force structure around large, exquisite platforms and then ask afterward how it can hide them. This makes deception more difficult because it is harder to design cheap decoys that effectively

¹ Eaglen, 2022; Geist, 2023.

² Horowitz, 2024.

mimic the signatures of exquisite platforms. The United States should also avoid applying a “gold-plating” mentality to decoys if it is possible to leverage AI and robotic mass for good-enough decoys to create uncertainty and confusion for enemy sensing systems.

The shift to deception will require the United States to invest in fog of war machines both as a metaphorical approach to warfighting and as a literal capability. The history of deception in war has been haphazard, with famous successes often depending on talented amateurs, but AI offers the opportunity to train a system that systematically optimizes creating persistent confusion and uncertainty. Because some deceptive tactics, such as camouflage and limiting movement or electronic emissions, impose a tax on U.S. forces, the fog of war machine can also treat deception as a constrained optimization problem to balance competing priorities of operational effectiveness and survivability. The fog of war machine will work best if it can orchestrate large numbers of physical and electronic decoys. Further analysis is required to figure out how to balance investments of scarce resources between new capabilities and new decoys.

The United States will need to counter adversary mass, not just field its own mass. In the long term, China could develop and field comparable or greater numbers of AI-enabled systems even if the United States started out with an advantage. In some cases, the United States might turn to symmetric responses, such as using drone swarms to defend against drone swarms. In other cases, it will need to invest in asymmetric responses that put it on the right side of the cost curve. The war in Ukraine has demonstrated that electronic warfare is one potential option: Ukraine reportedly lost 10,000 drones per month to Russian jamming in 2023.³ However, greater AI-enabled autonomy might reduce drones’ vulnerability to jamming. Another promising area for development is directed-energy weapons. Militaries have had long-standing hopes of employing directed-energy weapons, which are finally beginning to appear on mobile platforms on the battlefield.⁴ Further breakthroughs in directed-energy weapons could conceivably provide a more cost-effective way to counter smaller, unhardened systems. Although such breakthroughs are unlikely to stop the shift toward greater mass, they might limit the threat that the smallest and cheapest systems pose. AI is likely to play a role in managing multilayered air defense networks facing saturation attacks from an AI-enabled adversary.

The United States will face a growing challenge in defending the lines of communication required to project power and sustain a mass-based force structure over long distances. Deploying a much larger force structure would require developing new logistics capabilities to sustain that force. Logistical requirements will be an important factor informing what a new U.S. force structure might look like and how to mix expendable one-way drones that have low sustainment requirements with attritable systems that are cheaper and more numerous but still designed to be resupplied and reused over time. Sustainment challenges will be greatest when projecting power at great distances and when forces are widely distributed, especially because of the growing ability of U.S. competitors to conduct long-range precision strikes.⁵ Fielding autonomous logistics platforms at scale to sustain other uncrewed systems might be an important supporting capability.

The United States will need to continue making investments and reforms to increase the defense industrial base’s capacity to build a new force structure that prioritizes mass. The defense industrial base traditionally builds exquisite capabilities in small numbers, and it has recently struggled to do so. The United States is already working to strengthen the industrial base, and it should consider what kinds of targeted investments and reforms it might prioritize to help scale up the ability to generate robotic mass.

³ Watling, Danylyuk, and Reynolds, 2024.

⁴ See, for example, Venetia Gonzales, “Groundbreaking Laser Prototype Systems Delivered to 4-60th Air Defense Artillery Regiment,” U.S. Army, September 21, 2023.

⁵ Chris Smith, *The Implications of Emerging Changes in Land Warfare for the Focused All-Domain Defence Force*, Australian Strategic Policy Institute, December 2024, p. 19; Erik Sand, “Fighting to Supply the Fight—Assessing Approaches for Overcoming Contested Logistics,” *Naval War College Review*, Vol. 77, No. 2, Spring 2024.

For example, there are few of the large public shipyards required to build large, crewed warships, but building much smaller, uncrewed systems might make it possible to leverage smaller shipyards and thereby address one bottleneck that the Navy currently faces in increasing the size of its fleet. AI and robotic manufacturing might be part of the solution in establishing higher-volume production capacity and addressing bottlenecks from shortages of high-skilled human personnel.⁶ But many important changes will require reforms to bureaucratic processes, not just technological fixes.

Some of AI's most significant implications could lie in its effects on the underlying economics of warfare. Armies, navies, and air forces are exceedingly expensive. From the 18th through 20th centuries, technological advances in industrialization, metallurgy, synthetics, and mass production combined to change the capabilities and costs of weapons in ways that facilitated new tactics and force structures.⁷ In the 21st century, advances in AI—along with such supporting technologies as robotics and additive manufacturing—could do the same.⁸ Twenty-first-century weapon systems have hardware and software components, and AI could create major changes in the development and deployment cycle of such software. If AI succeeds in automating cognitive labor, there might also be large implications for military personnel management and the number of humans required to support military operations. Similarly, advances in robotics and declining costs could usher in a new relationship and division of labor between humans and machines in warfare. Although it is tempting to focus on exotic new capabilities, such as autonomous drone swarms, more-disruptive changes could come from cost reductions in producing and maintaining a range of new and old capabilities and how that shift changes the relative cost-effectiveness of different platforms and warfighting approaches. In sum, advanced AI is significant not just because it can replace humans for discrete tasks but because it can more fundamentally change what capabilities and approaches are cost-effective.

The advantage might go to the side that best executes the transition to an AI-enabled force. In addition to determining the right mix of capabilities, a key area for future research is how to manage the transition between the existing U.S. force structure and the one the U.S. military needs in order to win wars in the future. One long-standing concern that has inhibited military innovation in the past has been the fear that political authorities will call on militaries to fight when the militaries are in the middle of incorporating new innovations.⁹ The concern is that being caught during the transition is the worst of both worlds, when the military is neither prepared to fight in the way that had worked in the past nor ready to fight in the way it believes will be important in the future. Managing this transition requires a plan for addressing the organizational and cultural elements of adaptation, which includes developing thoughtful processes for how AI is fielded and integrated into units, monitoring how machines gain and maintain trust, and allowing variations in how units use AI-enabled capabilities.¹⁰ Successfully navigating the AI adoption process might be even more challenging and important than achieving the initial innovative breakthroughs with advanced AI, and it is far from clear that the United States will have an edge in the adoption competition.¹¹

⁶ Tucker, 2025.

⁷ William H. McNeill, *The Pursuit of Power: Technology, Armed Force, and Society Since A.D. 1000*, University of Chicago Press, 1982.

⁸ See also Jacquelyn Schneider and Julia Macdonald, “Looking Back to Look Forward: Autonomous Systems, Military Revolutions, and the Importance of Cost,” *Journal of Strategic Studies*, Vol. 47, No. 2, 2024.

⁹ Barry R. Posen, “Military Doctrine and the Management of Uncertainty,” *Journal of Strategic Studies*, Vol. 39, No. 2, 2016.

¹⁰ Jonathan P. Wong, Alexander C. Hou, Michael Miller, Katie A. Wilson, Emily Lathrop, Sydney Kessler, Sam Wallace, and Emily Yoder, *One Team One Fight: Vol. I, Insights on Human Machine Integration for the U.S. Army*, RAND Corporation, RR-A2764-1, 2025.

¹¹ Radha Iyengar Plumb and Michael C. Horowitz, “What America Gets Wrong About the AI Race: Winning Means Deploying, Not Just Developing, the Best Technology,” *Foreign Affairs*, April 18, 2025.

Remaining Uncertainties and Next Steps

In this report, we explore the potential consequences of technological breakthroughs that would allow AI to reliably and effectively substitute for human cognitive capacity in military operations. The predictions we make are necessarily tentative because AI is still maturing. However, AI's use in the war in Ukraine and recent USAF capability demonstrations suggest that the U.S. military does not need to wait until AGI is at hand to realize many of AI's benefits.

There are several areas of uncertainty that deserve further analysis in future research. First, we are more confident about the direction of the trends we discuss than the magnitude of the changes. For example, more-detailed analysis on the quantity versus quality competition is necessary to understand the size of the cost advantage of affordable mass over exquisite systems. The magnitude of the advantage might vary depending on assumptions about the different types of platforms, the requirements of different missions, and the attributes of an adversary's forces. This kind of detailed analysis would help translate our general prediction—that quantity will have a growing advantage—into specific recommendations about how many and what kinds of lower-cost systems the U.S. military should pursue.

Second, our analysis does not speak to speed and how long it will take for militaries to integrate AI into their forces and realize these advantages. There might be a striking difference in how militaries fight in the pre- and post-AI eras, but it might take many years or potentially even decades for militaries to reach the new equilibrium. The most-significant barriers to adoption could be organizational rather than technological. Developing a plan to overcome those organizational barriers requires further attention.

Third, our analysis highlights that AI's impact on military operations depends on its integration with supporting technologies. For example, AI's effect on the quantity-quality competition is mediated through its integration with uncrewed systems.¹² AI is, fundamentally, software, and its impact on the physical world depends on integrating it with hardware. Advances in other technologies could have important implications for the four building block competitions, such as quantum sensing for the hider-finder competition and directed energy for the quantity-quality competition. These supporting technologies deserve detailed monitoring and analysis. AI might be at the heart of integrating all these technological advances, but developing new operational concepts will require an understanding of the larger ecosystem of technologies and how those capabilities can all work together to create new and better ways of operating.

¹² Another important area for continuing study is these technologies' political effects on national security decisionmaking in addition to their operational effects on warfighting. See, for example, Erik Lin-Greenberg, *The Remote Revolution: Drones and Modern Statecraft*, Cornell University Press, 2025.

Estimating the Cost Ratio Between an Me 262 and a P-51D

In Chapter 2, we explored a historical example of quality versus quantity by comparing the German Me 262 fighter with the U.S. P-51D fighter. In discussing why quantity-based strategies can be more expensive than quality-based ones, we compared the cost of an Me 262 with the cost of a P-51D. In this appendix, we explain how we made that cost comparison.

We compared the relative costs by using cost-estimating relationship equations. Cost-estimating relationships use attributes of a system to estimate how much the system will cost. Cost-estimating relationships are developed through a statistical analysis of past production. The ratio we use is not an exact historical estimate of the unit price for each aircraft. It is a comparison of the costs that are required to build aircraft as capable as the Me 262 and the P-51D using their performance characteristics, holding all else equal. We used a RAND-developed cost-estimating relationship from 1972 in which the total cost is the sum of the engineering, development, tooling, flight-test operations, manufacturing labor, materials, and quality control costs.¹ We held constant the factors that were unrelated to aircraft performance, such as the labor costs, the total number of airframes produced, and the monthly production rate. The two parameters we varied were the maximum speeds and weights of the aircraft. We emphasize that this is not a historical cost estimate of the actual cost difference between the Me 262 and the P-51D. This is because the United States and Germany would have had different production costs, such as labor costs, that were unrelated to aircraft performance. But the cost-estimating relationship is still useful in providing a notional comparison of aircraft that have comparable differences in capabilities similar to those between the Me 262 and the P-51D.

Using this cost-estimating relationship, we estimated that an aircraft as capable as an Me 262 would cost about 1.5 times more than an aircraft as capable as a P-51D. If, in an aerial battle, the U.S. P-51Ds need to outnumber the German Me 262s by more than 3:1 to offset the 9:1 German lethality advantage (see Figure 2.2 in Chapter 2), then one P-51D would need to be less than one-third of the price of one Me 262 to come out cost neutral. But the price of one P-51D that we estimated was closer to 1/1.5 that of an Me 262. As a result, fielding the required number of P-51Ds to prevail over the Me 262s would be more expensive. For example, to win a battle against ten Me 262s, the United States would have to commit more than 30 P-51Ds, which would cost more than twice as much as Germany committed to its ten Me 262s. The calculation we used is as follows: $(30/10) \times (1/1.48) = 3/1.48 \sim 2$. This illustrates how quantity can overwhelm quality but at a high cost.

¹ G. S. Levenson, H. E. Boren, Dennis P. Tihansky, and Fred Timson, *Cost-Estimating Relationships for Aircraft Airframes*, RAND Corporation, R-761/1-ABR, 1972, pp. 53–56.

Abbreviations

AGI	artificial general intelligence
AI	artificial intelligence
ANI	artificial narrow intelligence
C2	command and control
CCA	collaborative combat aircraft
COA	course of action
DARPA	Defense Advanced Research Projects Agency
NP-hard	nondeterministic polynomial-time hard
UCAV	uncrewed combat aerial vehicle
USAF	U.S. Air Force

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